## Effective Radius of Curvature — Summary Table

| Geometry | Series | Name | R_apex (µm) | R_base (µm) | R_apex / d_cell | log10(R_base/R_apex) |
|---|---|---|---|---|---|---|
| **Ellipsoid — Aspect Ratio** | | | | | | |
| Ellipsoid | Aspect Ratio | 1:1 | 7.11 | 26.67 | 0.474 | 0.574 |
| Ellipsoid | Aspect Ratio | 2:1 | 3.62 | 29.09 | 0.241 | 0.905 |
| Ellipsoid | Aspect Ratio | 4:1 | 1.94 | 29.77 | 0.129 | 1.187 |
| **Ellipsoid — Scale** | | | | | | |
| Ellipsoid | Scale | 2:1 | 7.11 | 26.67 | 0.474 | 0.574 |
| Ellipsoid | Scale | 2:1 scaled 2x | 9.78 | 53.33 | 0.652 | 0.737 |
| Ellipsoid | Scale | 2:1 scaled 4x | 19.56 | 106.67 | 1.304 | 0.737 |
| **Hyperbolic Cone — Aspect Ratio** | | | | | | |
| Hyperbolic Cone | Aspect Ratio | 1:1 | 1.04* | 138.72 | 0.069 | 2.125 |
| Hyperbolic Cone | Aspect Ratio | 2:1 | 1.01* | 63.79 | 0.067 | 1.801 |
| Hyperbolic Cone | Aspect Ratio | 4:1 | 1.00* | 37.68 | 0.067 | 1.575 |
| **Hyperbolic Cone — Scale** | | | | | | |
| Hyperbolic Cone | Scale | 2:1 | 1.01* | 63.79 | 0.067 | 1.801 |
| Hyperbolic Cone | Scale | 2:1 scaled 2x | 1.01* | 123.51 | 0.067 | 2.087 |
| Hyperbolic Cone | Scale | 2:1 scaled 4x | 1.01* | 239.25 | 0.067 | 2.374 |

** Cone R_apex hits numerical r_min cutoff (geometric singularity, H → ∞). R_base = true max for ellipsoid; 95th percentile for cone (base approaches cylindrical, H → 0).*

| | | |
|---|---|---|
| Real Dimension | Height: 30 microns | |
| | | |
| | | |
| Hyperbolic 30H- Master Mold | Height | Diameter |
| 1 | 27.998 | 28.685 |
| 2 | 26.998 | 27.968 |
| 3 | 27.998 | 27.968 |
| 4 | 26.998 | 27.968 |
| Average | 27.498 | 28.14725 |
| STDEV | 0.5773502692 | 0.3585 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| MG_2_1: Feret's Diameter | | | | | | | |
| Semcircle | 32.919 | | | | | | |
| Cone | 29.708 | | | | | | |
| Hyperbolic | 29.961 | | average cell signal | 4.875 | | | |
| Triangular Prism | 28.617 | | | | | | |
| Triangular Pyramid | 26.664 | | background noise | 2.2 | | | |
| | | | | | | | |
| Semi | Intensity | signal to noise ratio | | | | | |
| 1 | 7.26667 | 3.303031818 | | | | | |
| 2 | 8.32812 | 3.785509091 | | | | | |
| 3 | 9.1802 | 4.172818182 | | | | | |
| 4 | | 0 | | | | | |
| | | | | | | | |
| | | | | | | | |
| Distance_(microns) | Gray_Value | | Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value |
| 0 | 3 | | 0 | 3.75 | | 0 | 3.25 |
| 0.9999 | 2.25833 | | 0.9999 | 3.9975 | | 0.9999 | 3.60355 |
| 1.9998 | 3 | | 1.9998 | 3.73595 | | 1.9998 | 3.10577 |
| 2.9997 | 3 | | 2.9997 | 3.56241 | | 2.9997 | 3.40902 |
| 3.9996 | 3 | | 3.9996 | 3.98373 | | 3.9996 | 3.66346 |
| 4.9995 | 3 | | 4.9995 | 4.38443 | | 4.9995 | 3.02515 |
| 5.9994 | 4 | | 5.9994 | 5.03698 | | 5.9994 | 3.06731 |
| 6.9993 | 4 | | 6.9993 | 3.4338 | | 6.9993 | 3.65533 |
| 7.9992 | 3 | | 7.9992 | 4.25 | | 7.9992 | 5.01433 |
| 8.9991 | 3.675 | | 8.999 | 6.15468 | | 8.999 | 5.30473 |
| 9.999 | 4 | | 9.9989 | 5.43861 | | 9.9989 | 6.24196 |
| 10.9989 | 4.34167 | | 10.9988 | 6.18981 | | 10.9988 | 7.75888 |
| 11.9988 | 5.65 | | 11.9987 | 7.8003 | | 11.9987 | 9.1802 |
| 12.9987 | 5.35833 | | 12.9986 | 8.32812 | | 12.9986 | 5.25 |
| 13.9986 | 7.26667 | | 13.9985 | 5.72189 | | 13.9985 | 0 |
| 14.9985 | 7.25 | | 14.9984 | 0.77385 | | 14.9984 | 0 |
| 15.9984 | 2.91667 | | 15.9983 | 0 | | 15.9983 | 0 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 16.9983 | 0 | | 16.9982 | 0 | | 16.9982 | 0 |
| 17.9982 | 0 | | 17.9981 | 0 | | 17.9981 | 0 |
| 18.9981 | 0 | | 18.998 | 0 | | 18.998 | 0 |
| 19.9981 | 0 | | 19.9979 | 0 | | 19.9979 | 0 |
| 20.998 | 0 | | 20.9978 | 0 | | 20.9978 | 0 |
| 21.9979 | 0 | | 21.9977 | 0 | | 21.9977 | 0 |
| 22.9978 | 0 | | 22.9976 | 0 | | 22.9976 | 0 |
| 23.9977 | 0 | | 23.9974 | 0 | | 23.9974 | 0 |
| 24.9976 | 0 | | 24.9973 | 0 | | 24.9973 | 0 |
| 25.9975 | 0 | | 25.9972 | 0 | | 25.9972 | 0 |
| 26.9974 | 0 | | | | | | |
| 27.9973 | 0 | | | | | | |
| 28.9972 | 0 | | | | | | |
| 29.9971 | 0 | | | | | | |
| | | | average cell signal | 4.102 | | | |
| | | | background noise | 2.2 | | | |
| Semi | Intensity | signal to noise ratio | | | | | |
| 1 | 5.16219 | 2.34645 | | | | | |
| 2 | 8.32812 | 3.785509091 | | | | | |
| 3 | 9.1802 | 4.172818182 | | | | | |
| | | | | | | | |
| | | | | | | | |
| Distance_(microns) | Gray_Value | | | | | | |
| 0 | 2.5 | | | | | | |
| 0.9997 | 3 | | | | | | |
| 1.9995 | 3.36829 | | | | | | |
| 2.9992 | 2.95351 | | | | | | |
| 3.999 | 2.88636 | | | | | | |
| 4.9987 | 3.16116 | | | | | | |
| 5.9985 | 3.78771 | | | | | | |
| 6.9982 | 4.75155 | | | | | | |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 7.998 | 5.33574 | | | | | | |
| 8.9977 | 5.16219 | | | | | | |
| 9.9975 | 7.11622 | | | | | | |
| 10.9972 | 6.3125 | | | | | | |
| 11.997 | 2.3595 | | | | | | |
| 12.9967 | 0.90909 | | | | | | |
| 13.9965 | 0.29442 | | | | | | |
| 14.9962 | 0 | | | | | | |
| 15.996 | 0 | | | | | | |
| 16.9957 | 0 | | | | | | |
| 17.9955 | 0 | | | | | | |
| 18.9952 | 0 | | | | | | |
| 19.995 | 0 | | | | | | |
| 20.9947 | 0 | | | | | | |
| 21.9945 | 0 | | | | | | |

| | | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 45 Degree | | | | | | | | | | | | | | | | | | | | | | | | | |
| | Area | Perimeter | Width | Height | Feret | | | | | | | | | | | | | | | | | | | | |
| 1 | 2594.821 | 239.875 | 12.916 | 10.999 | 48.996 | | | | | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1 | 1449.699 | 255 | 0 | 255 | 255 | 182.236 | 19.374 | 2 | 58.121 | 49.996 | 0 | 0 | 91.647 | 0.549 | 58.377 | 369673.22 | NaN | 343485 | 1 | 18 | 52 | 58.919 | 49.996 | NaN | Infinity | 0.934 |
| | | | | | | | | | | | | | | | | | | | | | | | | | |
| 1 | 711.396 | 255 | 0 | 255 | 255 | 127.871 | 7.534 | 6 | 25.832 | 45.997 | 0 | 0 | 90.254 | 0.547 | 48.078 | 181406.086 | NaN | 168555 | 1 | 7 | 52 | 73.08 | 25.779 | NaN | Infinity | 0.928 |
| | | | | | | | | | | | | | | | | | | | | | | | | | |
| | Top Width | Bottom Width | Height | | | | | | | | | | | | | | | | | | | | | | |
| | | | 52.33 | | | | | | | | | | | | | | | | | | | | | | |

| | | | | |
|---|---|---|---|---|
| | | | | |
| Turtle | Length | 80 | | |
| | Height: | 30 | | |
| | | | | |
| | | | | |
| Turtle _60H | | | | |
| | Length | 160 | L | |
| | Height: | 60 | Smaller height | 40 |
| | | | | |
| Exp | L | H1 | H2 | |
| 1 | 157.41 | 60.005 | 40.503 | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| Semi-Circle: Top | Intensity Lvl | Signal to noise ratio | | Background level noise | 1080.984 | | | | |
| 1 | 1192 | 1.102699022 | | | | | | Height: | |
| 2 | 5 | 0.004625415362 | | | | | | Semi-Circles | |
| 3 | 5 | 0.004625415362 | | | | | | 1 | 24.148 |
| 4 | 4.83333 | 0.004471231767 | | | | | | 2 | 14.25 |
| | | | | | | | | 3 | 17.25 |
| Distance_(micror | Gray_Value | | | | | | | 4 | 17 |
| 0 | 1026 | | | | | | | | |
| 0.16887 | 1037 | | | | | | | | |
| 0.33773 | 1055 | | | | | | | Cone | Height |
| 0.5066 | 1073 | | | | | | | 1 | 13 |
| 0.67547 | 1092 | | | | | | | 2 | 16.4 |
| 0.84433 | 1110 | | | | | | | 3 | 16 |
| 1.0132 | 1128 | | | | | | | 4 | 14.167 |
| 1.18207 | 1141 | | | | | | | | |
| 1.35093 | 1151 | | | | | | | Hyperbolics | |
| 1.5198 | 1162 | | | | | | | 1 | 12.25 |
| 1.68867 | 1172 | | | | | | | 2 | 11 |
| 1.85753 | 1182 | | | | | | | 3 | 13.5 |
| 2.0264 | 1192 | | | | | | | 4 | 10.25 |
| 2.19526 | 1192 | | | | | | | | |
| 2.36413 | 1188 | | | | | | | Pyramid | |
| 2.533 | 1185 | | | | | | | 1 | 14 |
| 2.70186 | 1181 | | | | | | | 2 | 15 |
| 2.87073 | 1177 | | | | | | | 3 | 12.25 |
| 3.0396 | 1174 | | | | | | | 4 | 13 |
| 3.20846 | 1171 | | | | | | | | |
| 3.37733 | 1169 | | | | | | | Prism | |
| 3.5462 | 1166 | | | | | | | 1 | 16.5 |
| 3.71506 | 1164 | | | | | | | 2 | 18.75 |
| 3.88393 | 1162 | | | | | | | 3 | 18.125 |
| 4.0528 | 1159 | | | | | | | 4 | 15.333 |
| 4.22166 | 1152 | | | | | | | | |
| 4.39053 | 1144 | | | | | | | | |
| 4.5594 | 1136 | | | | | | | | |
| 4.72826 | 1129 | | | | | | | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 4.89713 | 1121 | | | | | | | | |
| 5.06599 | 1113 | | | | | | | | |
| 5.23486 | 1115 | | | | | | | | |
| 5.40373 | 1116 | | | | | | | | |
| 5.57259 | 1118 | | | | | | | | |
| 5.74146 | 1119 | | | | | | | | |
| 5.91033 | 1121 | | | | | | | | |
| 6.07919 | 1121 | | | | | | | | |
| 6.24806 | 1114 | | | | | | | | |
| 6.41693 | 1106 | | | | | | | | |
| 6.58579 | 1098 | | | | | | | | |
| 6.75466 | 1090 | | | | | | | | |
| 6.92353 | 1083 | | | | | | | | |
| 7.09239 | 1074 | | | | | | | | |
| 7.26126 | 1063 | | | | | | | | |
| 7.43013 | 1052 | | | | | | | | |
| 7.59899 | 1041 | | | | | | | | |
| 7.76786 | 1030 | | | | | | | | |
| 7.93673 | 1019 | | | | | | | | |
| 8.10559 | 1013 | | | | | | | | |
| 8.27446 | 1028 | | | | | | | | |
| 8.44333 | 1042 | | | | | | | | |
| 8.61219 | 1057 | | | | | | | | |
| 8.78106 | 1071 | | | | | | | | |
| 8.94992 | 1086 | | | | | | | | |
| 9.11879 | 1094 | | | | | | | | |
| 9.28766 | 1087 | | | | | | | | |
| 9.45652 | 1080 | | | | | | | | |

| Semi-Circle: Top Signal | Intensity Lvl | Signal to noise ratio | | Background level noise | 2 | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 8.625 | 4.3125 | | | | | | | | | | | Height: | |
| 2 | 5 | 2.5 | | | | | | | | | | | Semi-Circles | |
| 3 | 5 | 2.5 | | | | | | | | | | | 1 | 14 |
| 4 | 4.83333 | 2.416665 | | | | | | | | | | | 2 | 14.25 |
| | | | | | | | | | | | | | 3 | 17.25 |
| | | | | | | | | | | | | | 4 | 17 |
| Selection | 1 | | Selection | 2 | | Selection | 3 | | | | | | Average | 15.625 |
| | | | | | | | | | | | | | STDEV | 1.738054468 |
| Distance_(microns) | Gray_Value | | Distance_(microns) | Gray_Value | | Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value | | | | |
| 0 | 2 | | 0 | 2 | | 0 | 0 | | 0 | 0 | | | Cone | Height |
| 1 | 2 | | 1 | 3 | | 1 | 0 | | 1 | 2 | | | 1 | 13 |
| 2 | 2 | | 2 | 2 | | 2 | 2 | | 2 | 2 | | | 2 | 16.4 |
| 3 | 2 | | 3 | 2 | | 3 | 2 | | 3 | 2 | | | 3 | 16 |
| 4 | 2.14815 | | 4 | 3 | | 4 | 2 | | 4 | 2 | | | 4 | 14.167 |
| | | | | | | | | | | | | | Average | 14.89175 |
| | | | | | | | | | | | | | STDEV | 1.59237943 |
| 5 | 2.64352 | | 5 | 3 | | 5 | 2 | | 5 | 2.0625 | | | | |
| 6 | 3.27778 | | 6 | 6 | | 6 | 2.07917 | | 6 | 3 | | | Hyperbolics | |
| 7 | 4.80556 | | 7 | 4 | | 7 | 3.04644 | | 7 | 3.10417 | | | 1 | 12.25 |
| 8 | 7.51852 | | 8 | 5 | | 8 | 4.70833 | | 8 | 4 | | | 2 | 11 |
| 9 | 8.625 | | 9 | 5 | | 9 | 4.625 | | 9 | 4.14523 | | | 3 | 13.5 |
| 10 | 6.89815 | | 10 | 5 | | 10 | 5 | | 10 | 4.83333 | | | 4 | 10.25 |
| | | | | | | | | | | | | | Average | 11.75 |
| | | | | | | | | | | | | | STDEV | 1.428869017 |
| 11 | 4.88426 | | 11 | 3 | | 11 | 2.76536 | | 11 | 3.81667 | | | | |
| 12 | 4.16667 | | 12 | 2 | | 12 | 2.58854 | | 12 | 3.0066 | | | Pyramid | |
| 13 | 3 | | 13 | 3 | | 13 | 3.25894 | | 13 | 3 | | | 1 | 14 |
| 14 | 3 | | 14 | 3 | | 14 | 3.23333 | | 14 | 3 | | | 2 | 15 |
| 15 | 3 | | 15 | 2 | | 15 | 2.64583 | | 15 | 2.73481 | | | 3 | 12.25 |
| 16 | 3 | | 16 | 2 | | 16 | 2.34757 | | 16 | 2.30938 | | | 4 | 13 |
| | | | | | | | | | | | | | Average | 13.5625 |
| | | | | | | | | | | | | | STDEV | 1.196783885 |
| 17 | 3 | | 17 | 2 | | 17 | 2 | | 17 | 3 | | | | |
| 18 | 2.91667 | | 18 | 2 | | 18 | 2 | | 18 | 2.66667 | | | Prism | |
| | | | 19 | 2 | | 19 | 2 | | 19 | 2 | | | 1 | 16.5 |
| | | | 20 | 2 | | 20 | 2.33854 | | 20 | 2 | | | 2 | 18.75 |
| | | | 21 | 2 | | | | | | | | | 3 | 18.125 |
| | | | 22 | 3 | | | | | | | | | 4 | 15.333 |
| | | | | | | | | | | | | | Average | 17.177 |
| | | | Peak Intensities: | 5 | | Peak | 4.777776667 | | | | | | STDEV | 1.552599755 |

| Cone: Top Signal | Intensity Lvl | Signal to noise ratio | | Background level noise | 2 |
|---|---|---|---|---|---|
| 1 | 6.93948 | 3.46974 | | | |
| 2 | N/A | #VALUE! | | | |
| 3 | 11.16667 | 5.583335 | | | |
| 4 | 5.75 | 2.875 | | | |
| Distance_(microns) | Gray_Value | | Distance_(microns) | Gray_Value | |
| 0 | 2 | | 0 | 2 | |
| 1 | 2 | | 1 | 2.02083 | |
| 2 | 2 | | 2 | 3.04167 | |
| 3 | 2 | | 3 | 4 | |
| 4 | 2 | | 4 | 3.91667 | |
| 5 | 2.04067 | | 5 | 3.3125 | |
| 6 | 2.33929 | | 6 | 6.625 | |
| 7 | 2.75 | | 7 | 10.41667 | |
| 8 | 3.0129 | | 8 | 7.5 | |

| Distance_(microns) | Gray_Value |
|---|---|
| 0 | 2.70312 |
| 1 | 2.02083 |
| 2 | 2.22396 |
| 3 | 2.09375 |
| 4 | 2.67708 |
| 5 | 2.375 |
| 6 | 3 |
| 7 | 3.52083 |
| 8 | 5.55729 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9 | 4.67857 | 9 | 10.375 | 9 | 5.75 | | |
| 10 | 6.93948 | 10 | 11.16667 | 10 | 5.47917 | | |
| 11 | 5.91171 | 11 | 8.45833 | 11 | 4.3125 | | |
| 12 | 3.56845 | 12 | 8.5 | 12 | 3.40625 | | |
| 13 | 3.75 | 13 | 4.27083 | 13 | 3.5 | | |
| 14 | 3.44444 | 14 | 5.29167 | 14 | 2.91146 | | |
| 15 | 3.07143 | 15 | 5.375 | 15 | 2.84375 | | |
| 16 | 3.15972 | 16 | 3.66667 | 16 | 3 | | |
| 17 | 2.26488 | 17 | 3 | 17 | 2.58333 | | |
| 18 | 2.75 | 18 | 2.625 | 18 | 2.4375 | | |
| 19 | 2.55754 | | | | | | |
| 20 | 2.4504 | | | | | | |
| 21 | 2.4375 | | | | | | |

| Cone: Top Signal | Intensity Lvl | Signal to noise ratio |
|---|---|---|
| 1 | 3.85507 | 1.927535 |
| 2 | 4.47917 | 2.239585 |
| 3 | 4.79167 | 2.395835 |
| 4 | 5.09954 | 2.54977 |

| Distance_(microns) | Gray_Value | Distance_(microns) | Gray_Value | Distance_(microns) | Gray_Value | Distance_(micror | Gray_Value |
|---|---|---|---|---|---|---|---|
| 0 | 2 | 0 | 2 | 0 | 2.27778 | 0 | 2 |
| 1 | 2 | 1 | 2.07639 | 1 | 2.56713 | 1 | 2 |
| 2 | 2 | 2 | 2.43056 | 2 | 2.05093 | 2 | 2.6713 |
| 3 | 2 | 3 | 2 | 3 | 2.11806 | 3 | 2.17361 |
| 4 | 2 | 4 | 2 | 4 | 2.50926 | 4 | 2 |
| 5 | 2.96377 | 5 | 2 | 5 | 3.39352 | 5 | 2 |
| 6 | 2.04348 | 6 | 2.125 | 6 | 2.66667 | 6 | 2 |
| 7 | 2.94928 | 7 | 3.43056 | 7 | 4.00926 | 7 | 2.12269 |
| 8 | 3 | 8 | 2.55556 | 8 | 3.83333 | 8 | 2.16667 |
| 9 | 2.06522 | 9 | 3.52083 | 9 | 4.79167 | 9 | 2.75694 |
| 10 | 3.85507 | 10 | 4.01389 | 10 | 3.90278 | 10 | 4.85185 |
| 11 | 3.07971 | 11 | 4.47917 | 11 | 3.43981 | 11 | 5.09954 |
| 12 | 3 | 12 | 3 | 12 | 3.41667 | 12 | 4.16667 |
| 13 | 3 | 13 | 3 | 13 | 4.71759 | 13 | 3.83333 |
| 14 | 3 | 14 | 3 | 14 | 4.74537 | 14 | 3.83333 |
| 15 | 3 | 15 | 3.95833 | 15 | 3.0625 | 15 | 3.3125 |
| 16 | 3 | 16 | 3.52778 | 16 | 2.55556 | 16 | 3 |
| 17 | 2.12319 | 17 | 3.03472 | 17 | 4.12268 | 17 | 3.09954 |
| 18 | 2.86957 | 18 | 3 | 18 | 2.22222 | 18 | 2.58333 |
| 19 | 2.13768 | 19 | 3.45139 | 19 | 2.92593 | 19 | 2.6412 |
| 20 | 2 | 20 | 2.61111 | 20 | 4.05556 | 20 | 2.44444 |
| 21 | 2.84783 | 21 | 2.5 | 21 | 2.20139 | 21 | 2.54167 |
| 22 | 3 | 22 | 2.5 | 22 | 2.97222 | 22 | 2.47222 |
| 23 | 2.16667 | 23 | 2.92361 | 23 | 2.00231 | 23 | 2.42824 |
| | | 24 | 2.16667 | 24 | 2.16667 | 24 | 2.83333 |

| Pyramid | Intensity Lvl | Signal to noise ratio |
|---|---|---|
| 1 | 4 | 2 |
| 2 | 3.71875 | 1.859375 |
| 3 | 5.24851 | 2.624255 |
| 4 | 4.98958 | 2.49479 |

| Distance_(microns) | Gray_Value | Distance_(microns) | Gray_Value | Distance_(microns) | Gray_Value | Distance_(micror | Gray_Value |
|---|---|---|---|---|---|---|---|
| 0 | 2 | 0 | 3 | 0 | 2.09375 | 0 | 2 |
| 1 | 2 | 1 | 2.03125 | 1 | 2.66518 | 1 | 2 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 2 | 2 | | 2 | 2 | 2 | 2.07589 | 2 | 2 | |
| 3 | 2 | | 3 | 2 | 3 | 2.68304 | 3 | 2 | |
| 4 | 2 | | 4 | 2 | 4 | 2 | 4 | 2 | |
| 5 | 3 | | 5 | 2 | 5 | 2 | 5 | 2 | |
| 6 | 3 | | 6 | 2.8125 | 6 | 2 | 6 | 2 | |
| 7 | 2 | | 7 | 3 | 7 | 2.01042 | 7 | 2.23698 | |
| 8 | 2 | | 8 | 3 | 8 | 2.27232 | 8 | 2.96875 | |
| 9 | 2 | | 9 | 3.71875 | 9 | 3.04018 | 9 | 2.27344 | |
| 10 | 3 | | 10 | 2.625 | 10 | 5.24851 | 10 | 2.00521 | |
| 11 | 4 | | 11 | 2 | 11 | 5.00149 | 11 | 2.98958 | |
| 12 | 3 | | 12 | 2 | 12 | 3.77232 | 12 | 3.75 | |
| 13 | 3 | | 13 | 2 | 13 | 3.02232 | 13 | 4.98958 | |
| 14 | 3 | | 14 | 2 | 14 | 3 | 14 | 4 | |
| 15 | 3 | | 15 | 2 | 15 | 3 | 15 | 3.96875 | |
| 16 | 2 | | 16 | 2 | 16 | 3 | 16 | 3 | |
| 17 | 3 | | | | 17 | 2.07738 | 17 | 2.94792 | |
| 18 | 2 | | | | 18 | 2.91071 | 18 | 2 | |
| 19 | 3 | | | | 19 | 2.10119 | 19 | 2 | |
| 20 | 2 | | | | 20 | 2 | 20 | 2.0625 | |
| 21 | 3 | | | | 21 | 2 | 21 | 2.70312 | |
| 22 | 2 | | | | | | 22 | 2.22396 | |
| 23 | 2 | | | | | | 23 | 2.08594 | |
| | | | | | | | 24 | 2.6875 | |

| Prism | Intensity Lvl | Signal to noise ratio |
|---|---|---|
| 1 | 5.48889 | 2.744445 |
| 2 | 4.25926 | 2.12963 |
| 3 | 5.6125 | 2.80625 |
| 4 | 5.66667 | 2.833335 |

| Distance_(microns) | Gray_Value | | Distance_(microns) | Gray_Value | | Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2 | | 0 | 2.11111 | | 0 | 2 | | 0 | 2.55556 |
| 1 | 2 | | 1 | 2.23148 | | 1 | 2 | | 1 | 2 |
| 2 | 2 | | 2 | 2.09259 | | 2 | 2 | | 2 | 2.05556 |
| 3 | 2 | | 3 | 2.33333 | | 3 | 2 | | 3 | 2.27778 |
| 4 | 2.42222 | | 4 | 2.40741 | | 4 | 2.45 | | 4 | 2.16667 |
| 5 | 3.37037 | | 5 | 2.60185 | | 5 | 3 | | 5 | 3.05556 |
| 6 | 4.37778 | | 6 | 2.44444 | | 6 | 3.425 | | 6 | 3.61111 |
| 7 | 5.48889 | | 7 | 3.18519 | | 7 | 4.4125 | | 7 | 5.11111 |
| 8 | 5.48889 | | 8 | 4.25926 | | 8 | 5.4 | | 8 | 5.66667 |
| 9 | 4.46667 | | 9 | 3.66667 | | 9 | 5.6125 | | 9 | 5.55556 |
| 10 | 4 | | 10 | 3.61111 | | 10 | 5 | | 10 | 4.94444 |
| 11 | 3.42222 | | 11 | 2.65741 | | 11 | 4.6375 | | 11 | 4.5 |
| 12 | 2.5 | | 12 | 2.33333 | | 12 | 3.65 | | 12 | 3.66667 |
| 13 | 2.68519 | | 13 | 3.62963 | | 13 | 3 | | 13 | 3.55556 |
| 14 | 2.35556 | | 14 | 2.72222 | | 14 | 3 | | 14 | 3.11111 |
| 15 | 2.55556 | | 15 | 2.05556 | | 15 | 2.6875 | | 15 | 3.5 |
| | | | 16 | 2.59259 | | 16 | 2.3 | | 16 | 2.72222 |
| | | | 17 | 2.09259 | | 17 | 2.7125 | | 17 | 3 |
| | | | 18 | 2.27778 | | 18 | 2.275 | | 18 | 3 |
| | | | | | | 19 | 3 | | 19 | 2.83333 |
| | | | | | | 20 | 2.75 | | | |

| Semi-Oval: Top Signal | Intensity Lvl | Signal to noise ratio | | Background level noise | 2 | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.46 | 2.73 | | | | | | | | | |
| 2 | 4.9999 | 2.49995 | | | | | | | | | |
| 3 | 5.5008 | 2.7504 | | | | | | | | | |
| 4 | 5.7791 | 2.88955 | | | | | | | | | |
| Average: | | | | | | | | | | | |
| | | | | | | | | | | | |
| Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value | | Distance_(micror | Gray_Value | | Distance_(micror | Gray_Value | |
| 0 | 2 | | 0 | 2 | | 0 | 2 | | 0 | 2 | |
| 0.5382 | 2 | | 0.5382 | 2 | | 0.5382 | 2 | | 0.5382 | 2 | |
| | | | | | | | | | | | |
| 1.0763 | 2 | | 1.0763 | 2 | | 1.0763 | 2 | | 1.0763 | 2 | |
| 1.6145 | 2 | | 1.6145 | 2 | | 1.6145 | 2 | | 1.6145 | 2 | |
| 2.1526 | 2 | | 2.1526 | 2 | | 2.1526 | 2 | | 2.1526 | 2 | |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| 2.6908 | 2 | | 2.6908 | 2.04 | | 2.6908 | 2 | | 2.6908 | 2 | |
| 3.229 | 2 | | 3.229 | 2.1539 | | 3.229 | 2 | | 3.229 | 2 | |
| 3.7671 | 2 | | 3.7671 | 2 | | 3.7671 | 2 | | 3.7671 | 2 | |
| 4.3053 | 2 | | 4.3053 | 2 | | 4.3053 | 2 | | 4.3053 | 2 | |
| 4.8434 | 2 | | 4.8434 | 2 | | 4.8434 | 2 | | 4.8434 | 2 | |
| 5.3816 | 2 | | 5.3816 | 2 | | 5.3816 | 2 | | 5.3816 | 2 | |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| 5.9198 | 2 | | 5.9198 | 2 | | 5.9198 | 2 | | 5.9198 | 2 | |
| 6.4579 | 2.0481 | | 6.4579 | 2 | | 6.4579 | 2.3969 | | 6.4579 | 2 | |
| 6.9961 | 2.38 | | 6.9961 | 2 | | 6.9961 | 2.62 | | 6.9961 | 2 | |
| 7.5342 | 2.3471 | | 7.5342 | 2 | | 7.5342 | 2.2379 | | 7.5342 | 2 | |
| 8.0724 | 2 | | 8.0724 | 2 | | 8.0724 | 2 | | 8.0724 | 2 | |
| 8.6106 | 2.0531 | | 8.6106 | 2 | | 8.6106 | 2 | | 8.6106 | 2 | |
| 9.1487 | 2.58 | | 9.1487 | 2 | | 9.1487 | 2.2436 | | 9.1487 | 2 | |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| 9.6869 | 2.6001 | | 9.6869 | 2 | | 9.6869 | 2.43 | | 9.6869 | 2 | |
| 10.225 | 3 | | 10.225 | 2.0636 | | 10.225 | 3 | | 10.225 | 2 | |
| 10.7632 | 2.995 | | 10.7632 | 3.1025 | | 10.7632 | 3.6975 | | 10.7632 | 2.3025 | |
| 11.3014 | 2.96 | | 11.3014 | 4.1168 | | 11.3014 | 4.46 | | 11.3014 | 2.54 | |
| 11.8395 | 3.9859 | | 11.8395 | 4.9991 | | 11.8395 | 5.2209 | | 11.8395 | 2.53 | |
| 12.3777 | 5.46 | | 12.3777 | 5 | | 12.3777 | 5.5008 | | 12.3777 | 2.7696 | |
| 12.9159 | 4.48 | | 12.9159 | 5.99 | | 12.9159 | 5.2499 | | 12.9159 | 3.51 | |
| 13.454 | 3.5 | | 13.454 | 6 | | 13.454 | 4.5 | | 13.454 | 4.25 | |
| 13.9922 | 4.5 | | 13.9922 | 4.9999 | | 13.9922 | 4.02 | | 13.9922 | 4.49 | |
| 14.5303 | 4.52 | | 14.5303 | 4.0196 | | 14.5303 | 3.8096 | | 14.5303 | 4.9792 | |
| 15.0685 | 3.56 | | 15.0685 | 2.1182 | | 15.0685 | 3.3127 | | 15.0685 | 5.7791 | |
| 15.6067 | 3.54 | | 15.6067 | 2.0416 | | 15.6067 | 2.7516 | | 15.6067 | 5.62 | |
| 16.1448 | 4.0725 | | 16.1448 | 2.0025 | | 16.1448 | 2.2475 | | 16.1448 | 4.8525 | |
| 16.683 | 4.12 | | 16.683 | 2 | | 16.683 | 2 | | 16.683 | 4.0672 | |
| 17.2211 | 3.4797 | | 17.2211 | 2 | | 17.2211 | 2 | | 17.2211 | 3.43 | |
| 17.7593 | 2.8864 | | 17.7593 | 2 | | 17.7593 | 2 | | 17.7593 | 3 | |
| 18.2975 | 2.82 | | 18.2975 | 2 | | 18.2975 | 2.4838 | | 18.2975 | 2.41 | |
| 18.8356 | 2.44 | | 18.8356 | 2 | | 18.8356 | 2.96 | | 18.8356 | 2.24 | |
| 19.3738 | 2.0429 | | 19.3738 | 2 | | 19.3738 | 2.3721 | | 19.3738 | 2 | |
| 19.9119 | 2.3344 | | 19.9119 | 2 | | 19.9119 | 2 | | 19.9119 | 2 | |
| 20.4501 | 2.37 | | 20.4501 | 2 | | 20.4501 | 2.1369 | | 20.4501 | 2 | |
| 20.9883 | 2.0504 | | 20.9883 | 2 | | 20.9883 | 2.36 | | 20.9883 | 2 | |
| 21.5264 | 2 | | 21.5264 | 2 | | 21.5264 | 2.2275 | | 21.5264 | 2 | |
| 22.0646 | 2.5544 | | 22.0646 | 2 | | 22.0646 | 2 | | 22.0646 | 2 | |
| 22.6027 | 2.67 | | 22.6027 | 2 | | 22.6027 | 2 | | 22.6027 | 2 | |
| 23.1409 | 2.1224 | | 23.1409 | 2 | | 23.1409 | 2 | | 23.1409 | 2 | |
| 23.6791 | 2 | | 23.6791 | 2 | | 23.6791 | 2 | | 23.6791 | 2 | |
| 24.2172 | 2 | | 24.2172 | 2 | | 24.2172 | 2 | | 24.2172 | 2 | |
| 24.7554 | 2 | | 24.7554 | 2 | | 24.7554 | 2 | | 24.7554 | 2 | |
| 25.2935 | 2.2184 | | 25.2935 | 2 | | 25.2935 | 2 | | 25.2935 | 2 | |
| 25.8317 | 2.27 | | 25.8317 | 2 | | 25.8317 | 2 | | 25.8317 | 2 | |
| 26.3699 | 2.0624 | | 26.3699 | 2 | | 26.3699 | 2.1924 | | 26.3699 | 2 | |
| 26.908 | 2 | | 26.908 | 2 | | 26.908 | 2.75 | | 26.908 | 2 | |
| | | | | | | | | | | | |
| | | | | | | | | | | | |
| Cone: Top Signal | Intensity Lvl | Signal to noise ratio | | Background level noise | 2 | | | | | | |
| 1 | 6.49545 | 3.247725 | | | | | | | | | |
| 2 | 3.47917 | 1.739585 | | | | | | | | | |
| 3 | 2.979167 | 1.4895835 | | | | | | | | | |
| 4 | 2.887 | 1.4435 | | | | | | | | | |
| | | | | | | | | | | | |
| | Distance_(micror | Gray_Value | | Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value | | Distance_(micror | Gray_Value |
| | 0 | 2 | | 0 | 2 | | 0 | 2 | | 0 | 2 |
| | 0.5382 | 2 | | 0.5382 | 2.24167 | | 0.5382 | 2 | | 0.5382 | 2 |
| | 1.0763 | 2 | | 1.0763 | 2.46667 | | 1.0763 | 2 | | 1.0763 | 2 |
| | 1.6145 | 2 | | 1.6145 | 2.225 | | 1.6145 | 2 | | 1.6145 | 2 |
| | 2.1526 | 2 | | 2.1526 | 2 | | 2.1526 | 2 | | 2.1526 | 2 |
| | 2.6908 | 2.10227 | | 2.6908 | 2 | | 2.6908 | 2 | | 2.6908 | 2 |
| | 3.229 | 2.19545 | | 3.229 | 2 | | 3.229 | 2 | | 3.229 | 2 |
| | 3.7671 | 2.09318 | | 3.7671 | 2 | | 3.7671 | 2 | | 3.7671 | 2 |
| | 4.3053 | 2 | | 4.3053 | 2 | | 4.3053 | 2 | | 4.3053 | 2 |
| | 4.8434 | 2 | | 4.8434 | 2 | | 4.8434 | 2.4375 | | 4.8434 | 2 |
| | 5.3816 | 2 | | 5.3816 | 2 | | 5.3816 | 2.88889 | | 5.3816 | 2 |
| | 5.9198 | 2 | | 5.9198 | 2 | | 5.9198 | 2.45139 | | 5.9198 | 2 |
| | 6.4579 | 2 | | 6.4579 | 2.5 | | 6.4579 | 2 | | 6.4579 | 2 |
| | 6.9961 | 2 | | 6.9961 | 3 | | 6.9961 | 2 | | 6.9961 | 2 |
| | 7.5342 | 2 | | 7.5342 | 2.63333 | | 7.5342 | 2.02778 | | 7.5342 | 2 |
| | 8.0724 | 2 | | 8.0724 | 2.625 | | 8.0724 | 2.04167 | | 8.0724 | 2 |
| | 8.6106 | 2 | | 8.6106 | 3.76667 | | 8.6106 | 2.01389 | | 8.6106 | 2 |
| | 9.1487 | 2 | | 9.1487 | 4.175 | | 9.1487 | 2 | | 9.1487 | 2 |
| | 9.6869 | 2.45682 | | 9.6869 | 2.9 | | 9.6869 | 3 | | 9.6869 | 2.275 |
| | 10.225 | 2.92273 | | 10.225 | 2 | | 10.225 | 3.47917 | | 10.225 | 3 |
| | 10.7632 | 2.5 | | 10.7632 | 2 | | 10.7632 | 2.5 | | 10.7632 | 3 |
| | 11.3014 | 2.08864 | | 11.3014 | 2.075 | | 11.3014 | 2.04167 | | 11.3014 | 2.88125 |
| | 11.8395 | 2.125 | | 11.8395 | 2.13333 | | 11.8395 | 2.05556 | | 11.8395 | 2.5 |

| Semi-Circles | Height |
|---|---|
| 1 | 28.253 |
| 2 | 29.331 |
| 3 | 30.137 |
| 4 | 26.37 |
| Average | 28.52275 |
| STDEV | 1.629536616 |

| Cone | Height |
|---|---|
| 1 | 25.024 |
| 2 | 27.044 |
| 3 | 26.908 |
| 4 | 27.984 |
| Average | 26.74 |
| STDEV | 1.240004301 |

| Hyperbolics | Height |
|---|---|
| 1 | 15.607 |
| 2 | 17.221 |
| 3 | 17.759 |
| 4 | 15.068 |
| Average | 16.41375 |
| STDEV | 1.281046285 |

| Prism | Height |
|---|---|
| 1 | 25.294 |
| 2 | 27.984 |
| 3 | 28.792 |
| 4 | 26.908 |
| Average | 27.2445 |
| STDEV | 1.512095566 |

| Pyramid | |
|---|---|
| 1 | 23.41 |
| 2 | 22.065 |
| 3 | 20.181 |
| 4 | 17.49 |
| AVerage | 20.7865 |
| STDEV | 2.565856387 |

Chip

HemiEllipsoids

| | Base | Transition | Tip | nucleus | height | nuc/base | nuc/height | reach at top | over? |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.356666667 | 7.222333333 | 2.761 | 35.1 | 35.425 | 4.771182601 | 99.08256881 | yes | no |
| 2 | 7.096 | 6.075666667 | 2.768 | 33.963 | 37.05 | 4.786217587 | 91.66801619 | yes | yes |
| 3 | 6.873 | 6.616166667 | 3.239 | 33.15 | 36.075 | 4.823221301 | 91.89189189 | yes | yes |
| 4 | 7.01 | 7.697 | 5.259 | 37.375 | 37.4 | 5.331669044 | 99.93315508 | yes | yes |
| 5 | 6.338 | 5.542333333 | 2.625333333 | 30.716 | 36.888 | 4.846323761 | 83.26827152 | yes | yes |
| AVG | 6.934733333 | 6.6307 | 3.330466667 | 34.0608 | 36.5676 | 4.911722859 | 93.1687807 | | |
| standard dev | 0.3773597164 | 0.8631107853 | 1.10289177 | 2.454112202 | 0.802565449 | 0.2366260254 | 6.757079095 | | |

Cone

| | Base | Transition | Tip | nucleus | height | nuc/base | Nuc/Height % | Reach at top | over? |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.991 | 6.854 | 1.15 | 10.4 | 34.613 | 1.487626949 | 30.04651432 | yes | no |
| 2 | 7.205333333 | 6.006666667 | 1.35 | 16.088 | 33.963 | 2.232790526 | 47.36919589 | yes | yes |
| 3 | 6.283666667 | 6.095333333 | 1.492766667 | 18.85 | 32.175 | 2.999840857 | 58.58585859 | yes | yes |
| 4 | 7.312666667 | 5.607 | 2.20925 | 20.313 | 33.8 | 2.777782843 | 60.09763314 | yes | yes |
| 5 | 7.367 | 6.053333333 | 1.397333333 | 16.9 | 34.775 | 2.294013846 | 48.59813084 | no | no |
| AVG | 7.031933333 | 6.123266667 | 1.51987 | 16.5102 | 33.8652 | 2.358411004 | 48.93946655 | | |
| standard dev | 0.4423592934 | 0.4527054721 | 0.4051710267 | 3.794756013 | 1.031689488 | 0.5842724888 | | | |

Hyperbolics

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.731 | 5.677 | 0.998 | 10.239 | 24.7 | 1.324408227 | 41.4534413 | No | No |
| 2 | 6.554 | 7.0135 | 1.425 | 8.775 | 28.275 | 1.338877022 | 31.03448276 | No | No |
| 3 | 6.988 | 7.509 | 0.711 | 9.263 | 27.625 | 1.3255581 | 33.53122172 | Yes | Yes |
| 4 | 4.257 | 5.17 | 2.153 | 11.7 | 25.838 | 2.748414376 | 45.28214258 | Yes | Yes |
| 5 | 6.326 | 6.832 | 1.3945 | 10.238 | 23.486 | 1.618400253 | 43.59192711 | yes | Yes |
| AVG= | 6.3712 | 6.4403 | 1.3363 | 10.043 | 25.9848 | 1.671131595 | 38.97864309 | | |
| Standard Dev | 1.297574545 | 0.9772916402 | 0.5439900734 | 1.122309449 | 1.990699299 | 0.615091461 | 6.323076869 | | |

Triangle Prism

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 9.48 | 8.812333333 | 3.505666667 | 29.738 | 33.963 | 3.136919831 | 87.55999176 | yes | no |
| 2 | 4.172 | 5.824 | 4.313333333 | 17.226 | 34.775 | 4.128954938 | 49.53558591 | yes | YES |
| 3 | 4.712666667 | 5.212333333 | 4.742666667 | 32.175 | 35.75 | 6.827344745 | 90 | yes | yes |
| 4 | 5.579333333 | 9.340666667 | 3.125 | 20.313 | 34.125 | 3.640757558 | 59.52527473 | yes | no |
| 5 | 8.179333333 | 7.060666667 | 3.091333333 | 21.938 | 35.1 | 2.682125683 | 62.5014245 | yes | no |
| AVG= | 6.424666667 | 7.25 | 3.7556 | 24.278 | 34.7426 | 4.083220551 | 69.82445538 | | |
| Standard Dev | 2.298584154 | 1.805074868 | 0.7393815509 | 6.38556454 | 0.730213188 | 1.626855449 | 17.9787373 | | |

Square Pyramid

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.991333333 | 4.456333333 | 2.494 | 24.863 | 30.875 | 3.556260132 | 80.52793522 | No | No |
| 2 | 7.779333333 | 7.044 | 2.109333333 | 23.077 | 32.313 | 2.966449567 | 71.41707672 | No | No |
| 3 | 5.788333333 | 7.628666667 | 2.421 | 21.135 | 31.204 | 3.651310107 | 67.73170106 | Yes | Yes |
| 4 | 7.535333333 | 4.034 | 1.459333333 | 11.05 | 32.988 | 1.466424843 | 33.49702922 | Yes | Yes |
| 5 | 6.068 | 5.476666667 | 1.924666667 | 22.1 | 29.575 | 3.642056691 | 74.72527473 | yes | Yes |
| AVG= | 6.832466667 | 5.727933333 | 2.081666667 | 20.445 | 31.391 | 3.056500268 | 65.57980339 | | |
| Standard Dev | 0.8789824167 | 1.572762721 | 0.4176957957 | 5.429716797 | 1.322835402 | 0.9331243186 | 18.54113746 | | |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 12.3777 | 2.58182 | | 12.3777 | 2.05833 | | 12.3777 | 2.06944 | | 12.3777 | 2.5 |
| | 12.9159 | 3.5 | | 12.9159 | 2 | | 12.9159 | 2.04167 | | 12.9159 | 2.5 |
| | 13.454 | 4.47727 | | 13.454 | 2.04167 | | 13.454 | 2.45139 | | 13.454 | 2.5 |
| | 13.9922 | 5.98636 | | 13.9922 | 2.06667 | | 13.9922 | 2.88889 | | 13.9922 | 2.5 |
| | 14.5303 | 6.49545 | | 14.5303 | 2.025 | | 14.5303 | 2.4375 | | 14.5303 | 2.5 |
| | 15.0685 | 5.49318 | | 15.0685 | 2.01667 | | 15.0685 | 2 | | 15.0685 | 2.5 |
| | 15.6067 | 4.49318 | | 15.6067 | 2.51667 | | 15.6067 | 2 | | 15.6067 | 2.43125 |
| | 16.1448 | 4 | | 16.1448 | 3 | | 16.1448 | 2 | | 16.1448 | 2 |
| | 16.683 | 3.5 | | | | | 16.683 | 2 | | 16.683 | 2 |
| | 17.2211 | 3 | | | | | 17.2211 | 2 | | 17.2211 | 2 |
| | 17.7593 | 3.45 | | | | | 17.7593 | 2 | | 17.7593 | 2 |
| | 18.2975 | 3.88182 | | | | | 18.2975 | 2 | | 18.2975 | 2 |
| | 18.8356 | 3.86364 | | | | | 18.8356 | 2 | | 18.8356 | 2 |
| | 19.3738 | 3.38409 | | | | | 19.3738 | 2 | | 19.3738 | 2 |
| | 19.9119 | 2.45682 | | | | | | | | 19.9119 | 2 |
| | 20.4501 | 2 | | | | | | | | 20.4501 | 2 |
| | 20.9883 | 2.44773 | | | | | | | | 20.9883 | 2 |
| | 21.5264 | 2.5 | | | | | | | | 21.5264 | 2 |
| | 22.0646 | 2.12273 | | | | | | | | | |
| | 22.6027 | 2.06591 | | | | | | | | | |
| | 23.1409 | 2 | | | | | | | | | |
| | 23.6791 | 2 | | | | | | | | | |
| | 24.2172 | 2 | | | | | | | | | |
| | 24.7554 | 2 | | | | | | | | | |
| | 25.2935 | 2 | | | | | | | | | |
| | 25.8317 | 2.09318 | | | | | | | | | |
| | 26.3699 | 2.19545 | | | | | | | | | |
| | 26.908 | 2.10227 | | | | | | | | | |
| | 27.4462 | 2 | | | | | | | | | |
| | 27.9843 | 2 | | | | | | | | | |
| | 28.5225 | 2 | | | | | | | | | |
| | 29.0607 | 2 | | | | | | | | | |
| | 29.5988 | 2 | | | | | | | | | |

| Cone: Top Signal | Intensity Lvl | Signal to noise ratio | | Background level noise | 2 |
|---|---|---|---|---|---|
| 1 | 3.82258 | 1.91129 | | | |
| 2 | 3.85484 | 1.92742 | | | |
| 3 | 3.48784 | 1.74392 | | | |
| 4 | 2.895161 | 1.4475805 | | | |

| Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value | | Distance_(micror | Gray_Value | | Distance_(micror | Gray_Value |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2 | | 0 | 2 | | 0 | 2 | | 0 | 2 |
| 0.5382 | 2 | | 0.5382 | 2.17163 | | 0.5382 | 2 | | 0.5382 | 2 |
| 1.0763 | 2 | | 1.0763 | 2.21774 | | 1.0763 | 2 | | 1.0763 | 2 |
| 1.6145 | 2 | | 1.6145 | 2.06016 | | 1.6145 | 2 | | 1.6145 | 2 |
| 2.1526 | 2.18965 | | 2.1526 | 2 | | 2.1526 | 2 | | 2.1526 | 2 |
| 2.6908 | 2.41935 | | 2.6908 | 2 | | 2.6908 | 2 | | 2.6908 | 2 |
| 3.229 | 2.24063 | | 3.229 | 2 | | 3.229 | 2 | | 3.229 | 2 |
| 3.7671 | 2 | | 3.7671 | 2 | | 3.7671 | 2.04975 | | 3.7671 | 2 |
| 4.3053 | 2 | | 4.3053 | 2 | | 4.3053 | 2.37903 | | 4.3053 | 2 |
| 4.8434 | 2 | | 4.8434 | 2 | | 4.8434 | 2.35373 | | 4.8434 | 2 |
| 5.3816 | 2 | | 5.3816 | 3.07297 | | 5.3816 | 2.03649 | | 5.3816 | 2 |
| 5.9198 | 2 | | 5.9198 | 3.85484 | | 5.9198 | 2.42742 | | 5.9198 | 2 |
| 6.4579 | 2 | | 6.4579 | 3.36206 | | 6.4579 | 2.47496 | | 6.4579 | 2 |
| 6.9961 | 2.20604 | | 6.9961 | 2.44114 | | 6.9961 | 3.02179 | | 6.9961 | 2 |
| 7.5342 | 2.72581 | | 7.5342 | 2 | | 7.5342 | 3.52302 | | 7.5342 | 2 |
| 8.0724 | 3 | | 8.0724 | 2 | | 8.0724 | 3.48784 | | 8.0724 | 2 |
| 8.6106 | 3.57466 | | 8.6106 | 2 | | 8.6106 | 3.0201 | | 8.6106 | 2 |
| 9.1487 | 2.82518 | | 9.1487 | 2 | | 9.1487 | 2.53688 | | 9.1487 | 2.011512 |
| 9.6869 | 2.37539 | | 9.6869 | 2.01854 | | 9.6869 | 2.54032 | | 9.6869 | 2.040323 |
| 10.225 | 3.19355 | | 10.225 | 2.47496 | | 10.225 | 2.03141 | | 10.225 | 2.031413 |
| 10.7632 | 3.82258 | | 10.7632 | 3 | | 10.7632 | 2 | | 10.7632 | 2 |
| 11.3014 | 2.97399 | | 11.3014 | 2.96351 | | 11.3014 | 2.53649 | | 11.3014 | 2.374805 |
| 11.8395 | 2.73075 | | 11.8395 | 2.89516 | | 11.8395 | 2.41714 | | 11.8395 | 2.895161 |
| 12.3777 | 2 | | 12.3777 | 2.54585 | | 12.3777 | 2.37903 | | 12.3777 | 2.879032 |
| 12.9159 | 2 | | 12.9159 | 2 | | 12.9159 | 2.04975 | | 12.9159 | 2.862903 |
| 13.454 | 2 | | 13.454 | 2 | | 13.454 | 2 | | 13.454 | 2.846774 |
| 13.9922 | 2.07414 | | 13.9922 | 2 | | 13.9922 | 2 | | 13.9922 | 2.830645 |
| 14.5303 | 2.93548 | | 14.5303 | 2 | | 14.5303 | 2 | | 14.5303 | 2.558338 |
| 15.0685 | 2.90557 | | 15.0685 | 2 | | 15.0685 | 3.12032 | | 15.0685 | 2 |
| 15.6067 | 2 | | 15.6067 | 2 | | 15.6067 | 3.43548 | | 15.6067 | 2 |
| 16.1448 | 2 | | 16.1448 | 2 | | 16.1448 | 3.67163 | | 16.1448 | 2 |
| 16.683 | 2 | | 16.683 | 2 | | 16.683 | 3.1875 | | 16.683 | 2 |

| Prism: Top Signal | Intensity Lvl | Signal to noise ratio | | Background level noise | 2 |
|---|---|---|---|---|---|
| 1 | 3.5 | 1.75 | | | |
| 2 | 4 | 2 | | | |
| 3 | 7.5 | 3.75 | | | |
| 4 | 3.80769 | 1.903845 | | | |

| | Distance_(micror | Gray_Value | | Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value | Distance_(micror | Gray_Value |
|---|---|---|---|---|---|---|---|---|---|---|
| | 0 | 2 | | 0 | 2 | | 0 | 2.0625 | 0 | 2 |
| | 0.5382 | 2 | | 0.53816 | 2 | | 0.5382 | 2 | 0.5382 | 2 |
| | 1.0763 | 2.25 | | 1.07632 | 2 | | 1.0763 | 2 | 1.0763 | 2 |
| | 1.6145 | 2.5 | | 1.61448 | 2 | | 1.6145 | 2 | 1.6145 | 2.41679 |
| | 2.1526 | 2.25 | | 2.15264 | 2 | | 2.1526 | 2 | 2.1526 | 2.92308 |
| | 2.6908 | 2 | | 2.6908 | 3 | | 2.6908 | 2.16518 | 2.6908 | 2.53883 |
| | 3.229 | 2 | | 3.22896 | 4 | | 3.229 | 2.25 | 3.229 | 2 |
| | 3.7671 | 2 | | 3.76712 | 3 | | 3.7671 | 2.5625 | 3.7671 | 2.04919 |
| | 4.3053 | 2 | | 4.30528 | 2 | | 4.3053 | 2.75 | 4.3053 | 2.44675 |
| | 4.8434 | 2.25 | | 4.84344 | 2 | | 4.8434 | 3.33929 | 4.8434 | 3.27034 |
| | 5.3816 | 2.5 | | 5.3816 | 2.333333 | | 5.3816 | 3.75 | 5.3816 | 3.80769 |
| | 5.9198 | 2.5 | | 5.91976 | 2.266667 | | 5.9198 | 3.33482 | 5.9198 | 3.56102 |
| | 6.4579 | 2.5 | | 6.45793 | 2.2 | | 6.4579 | 4.07143 | 6.4579 | 3 |
| | 6.9961 | 2.25 | | 6.99609 | 2.133333 | | 6.9961 | 6.03571 | 6.9961 | 3 |
| | 7.5342 | 2 | | 7.53425 | 2.066667 | | 7.5342 | 7.25 | 7.5342 | 3.16864 |
| | 8.0724 | 2.5 | | 8.07241 | 2 | | 8.0724 | 7.5 | 8.0724 | 3.71154 |

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 8.6106 | 3 | | | | | 8.6106 | 6.45536 | 8.6106 | 3.5 |
| | 9.1487 | 3.25 | | | | | 9.1487 | 4.69196 | 9.1487 | 2.55658 |
| | 9.6869 | 3.5 | | | | | 9.6869 | 4 | 9.6869 | 2 |
| | 10.225 | 3.5 | | | | | 10.225 | 4.41071 | 10.225 | 2 |
| | 10.7632 | 3.5 | | | | | 10.7632 | 5.39286 | 10.7632 | 2 |
| | 11.3014 | 3.25 | | | | | 11.3014 | 6.09375 | 11.3014 | 2.03883 |
| | 11.8395 | 2.75 | | | | | 11.8395 | 6.07143 | 11.8395 | 2.5 |
| | 12.3777 | 2.5 | | | | | 12.3777 | 5.66518 | 12.3777 | 3.41679 |
| | 12.9159 | 2.25 | | | | | 12.9159 | 5.17857 | 12.9159 | 2.96154 |
| | 13.454 | 2 | | | | | 13.454 | 4.5 | 13.454 | 2 |
| | 13.9922 | 2 | | | | | 13.9922 | 4.71429 | 13.9922 | 2 |
| | 14.5303 | 2 | | | | | 14.5303 | 5.11607 | | |
| | 15.0685 | 2 | | | | | 15.0685 | 4.6875 | | |
| | 15.6067 | 2 | | | | | | | | |
| | 16.1448 | 2 | | | | | | | | |
| | 16.683 | 2 | | | | | | | | |
| | 17.2211 | 2 | | | | | | | | |
| | 17.7593 | 2 | | | | | | | | |
| | 18.2975 | 2 | | | | | | | | |
| | 18.8356 | 2 | | | | | | | | |
| | 19.3738 | 2 | | | | | | | | |
| | 19.9119 | 2 | | | | | | | | |
| | 20.4501 | 2 | | | | | | | | |
| | 20.9883 | 2.25 | | | | | | | | |
| | 21.5264 | 2.5 | | | | | | | | |
| | 22.0646 | 2.75 | | | | | | | | |
| | 22.6027 | 3 | | | | | | | | |
| | 23.1409 | 2.5 | | | | | | | | |
| | 23.6791 | 2 | | | | | | | | |
| | 24.2172 | 2.25 | | | | | | | | |
| | 24.7554 | 2.25 | | | | | | | | |
| | 25.2935 | 2 | | | | | | | | |

| Prism: Top Signal | Intensity Lvl | Signal to noise ratio | | Background level noise | 2 |
|---|---|---|---|---|---|
| 1 | 3.5 | 1.75 | | | |
| 2 | 4.5 | 2.25 | | | |
| 3 | 4.125 | 2.0625 | | | |
| 4 | 3.7907 | 1.89535 | | | |

| Distance_(microns) | Gray_Value | | Distance_(micror | Gray_Value | | Distance_(micror | Gray_Value | | Distance_(micror | Gray_Value |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2 | | 0 | 2 | | 0 | 2.1875 | | 0 | 2 |
| 0.5382 | 2 | | 0.5382 | 2 | | 0.5382 | 2 | | 0.5382 | 2 |
| 1.0763 | 2 | | 1.0763 | 2 | | 1.0763 | 2 | | 1.0763 | 2 |
| 1.6145 | 2 | | 1.6145 | 2 | | 1.6145 | 2 | | 1.6145 | 2 |
| 2.1526 | 2 | | 2.1526 | 2 | | 2.1526 | 2 | | 2.1526 | 2 |
| 2.6908 | 2 | | 2.6908 | 2 | | 2.6908 | 2 | | 2.6908 | 2 |
| 3.229 | 2 | | 3.229 | 2 | | 3.229 | 2 | | 3.229 | 2 |
| 3.7671 | 2 | | 3.7671 | 2 | | 3.7671 | 2 | | 3.7671 | 2 |
| 4.3053 | 2 | | 4.3053 | 2 | | 4.3053 | 2 | | 4.3053 | 2 |
| 4.8434 | 2 | | 4.8434 | 2 | | 4.8434 | 2.10938 | | 4.8434 | 2 |
| 5.3816 | 3.5 | | 5.3816 | 2 | | 5.3816 | 3.05566 | | 5.3816 | 2 |
| 5.9198 | 3.5 | | 5.9198 | 2 | | 5.9198 | 3.62329 | | 5.9198 | 2.04759 |
| 6.4579 | 2.5 | | 6.4579 | 2 | | 6.4579 | 4.125 | | 6.4579 | 2.13953 |
| 6.9961 | 2 | | 6.9961 | 2.0603 | | 6.9961 | 4.08936 | | 6.9961 | 2.09843 |
| 7.5342 | 2 | | 7.5342 | 2.28027 | | 7.5342 | 3.9834 | | 7.5342 | 2 |
| 8.0724 | 2 | | 8.0724 | 2.40649 | | 8.0724 | 3.47632 | | 8.0724 | 2 |
| 8.6106 | 2 | | 8.6106 | 2.1875 | | 8.6106 | 3 | | 8.6106 | 2.25554 |
| 9.1487 | 2 | | 9.1487 | 2.23438 | | 9.1487 | 2.52368 | | 9.1487 | 3.10465 |
| 9.6869 | 2 | | 9.6869 | 3.06152 | | 9.6869 | 2.4707 | | 9.6869 | 3.7907 |
| 10.225 | 2 | | 10.225 | 3.29688 | | 10.225 | 2.54248 | | 10.225 | 3.62331 |
| 10.7632 | 2.5 | | 10.7632 | 3.44141 | | 10.7632 | 2.5625 | | 10.7632 | 3.23256 |
| 11.3014 | 2.5 | | 11.3014 | 4.11548 | | 11.3014 | 2.04517 | | 11.3014 | 3.24419 |
| 11.8395 | 2.5 | | 11.8395 | 4.5 | | 11.8395 | 2 | | 11.8395 | 3.01163 |
| 12.3777 | 2 | | 12.3777 | 3.5 | | 12.3777 | 2.3479 | | 12.3777 | 2.26744 |
| | | | 12.9159 | 2.625 | | 12.9159 | 2.375 | | 12.9159 | 2.5 |
| | | | 13.454 | 2.58545 | | 13.454 | 2.60107 | | 13.454 | 3.43916 |
| | | | 13.9922 | 2.42383 | | 13.9922 | 2.65625 | | 13.9922 | 3.80233 |
| | | | 14.5303 | 2.57812 | | 14.5303 | 2.67188 | | 14.5303 | 3 |
| | | | 15.0685 | 2.58984 | | 15.0685 | 2.6875 | | 15.0685 | 2.94321 |
| | | | 15.6067 | 3 | | 15.6067 | 2.14282 | | 15.6067 | 2.71769 |
| | | | 16.1448 | 2.9834 | | 16.1448 | 2 | | 16.1448 | 3.05273 |
| | | | 16.683 | 2.48438 | | 16.683 | 2 | | 16.683 | 3.36047 |
| | | | 17.2211 | 2.5 | | 17.2211 | 2 | | 17.2211 | 3.24419 |
| | | | | | | | | | 17.7593 | 2.38372 |
| | | | | | | | | | 18.2975 | 2.39535 |
| | | | | | | | | | 18.8356 | 2.46214 |
| | | | | | | | | | 19.3738 | 3 |
| | | | | | | | | | 19.9119 | 2.93023 |
| | | | | | | | | | 20.4501 | 2 |
| | | | | | | | | | 20.9883 | 2.02542 |
| | | | | | | | | | 21.5264 | 2.53488 |
| | | | | | | | | | 22.0646 | 2.52217 |
| | | | | | | | | | 22.6027 | 2.48837 |
| | | | | | | | | | 23.1409 | 2.5 |

2:1

HemiEllipsoids

| | Base | Transition | Tip | nucleus | height | nuc/base | nuc/height | reach at top | over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.356666667 | 7.222333333 | 2.761 | 35.1 | 35.425 | 4.771182601 | 99.08256881 | yes | no | 4 |
| 2 | 7.096 | 6.075666667 | 2.768 | 33.963 | 37.05 | 4.786217587 | 91.66801619 | yes | yes | 3 |
| 3 | 6.873 | 6.616166667 | 3.239 | 33.15 | 36.075 | 4.823221301 | 91.89189189 | yes | yes | 6 |
| 4 | 7.01 | 7.697 | 5.259 | 37.375 | 37.4 | 5.331669044 | 99.93315508 | yes | yes | 5 |
| 5 | 6.338 | 5.542333333 | 2.625333333 | 30.716 | 36.888 | 4.846323761 | 83.26827152 | yes | yes | 4 |
| AVG | 6.934733333 | 6.6307 | 3.330466667 | 34.0608 | 36.5676 | 4.911722859 | 93.1687807 | 5/5 | 4/5 | 4.4 |
| standard dev | 0.3773597164 | 0.8631107853 | 1.10289177 | 2.454112202 | 0.802565449 | 0.2366260254 | 6.757079095 | | | 1.140175425 |

Cone

| | Base | Transition | Tip | nucleus | height | nuc/base | Nuc/Height % | Reach at top | over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.991 | 6.854 | 1.15 | 10.4 | 34.613 | 1.487626949 | 30.04651432 | yes | no | 2 |
| 2 | 7.205333333 | 6.006666667 | 1.35 | 16.088 | 33.963 | 2.232790526 | 47.36919589 | yes | yes | 1 |
| 3 | 6.283666667 | 6.095333333 | 1.492766667 | 18.85 | 32.175 | 2.999840857 | 58.58585859 | yes | yes | 3 |
| 4 | 7.312666667 | 5.607 | 2.20925 | 20.313 | 33.8 | 2.777782843 | 60.09763314 | yes | yes | 3 |
| 5 | 7.367 | 6.053333333 | 1.397333333 | 16.9 | 34.775 | 2.294013846 | 48.59813084 | no | no | 2 |
| AVG | 7.031933333 | 6.123266667 | 1.51987 | 16.5102 | 33.8652 | 2.358411004 | 48.93946655 | 4/5 | 3/5 | 2.2 |
| standard dev | 0.4423592934 | 0.4527054721 | 0.4051710267 | 3.794756013 | 1.031689488 | 0.5842724888 | 12.01128079 | | | 0.8366600265 |

Hyperbolics

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.731 | 5.677 | 0.998 | 10.239 | 24.7 | 1.324408227 | 41.4534413 | No | No | 1 |
| 2 | 6.554 | 7.0135 | 1.425 | 8.775 | 28.275 | 1.338877022 | 31.03448276 | No | No | 1 |
| 3 | 6.988 | 7.509 | 0.711 | 9.263 | 27.625 | 1.3255581 | 33.53122172 | Yes | Yes | 2 |
| 4 | 4.257 | 5.17 | 2.153 | 11.7 | 25.838 | 2.748414376 | 45.28214258 | Yes | Yes | 2 |
| 5 | 6.326 | 6.832 | 1.3945 | 10.238 | 23.486 | 1.618400253 | 43.59192711 | yes | Yes | 1 |
| AVG= | 6.3712 | 6.4403 | 1.3363 | 10.043 | 25.9848 | 1.671131595 | 38.97864309 | 3/5 | 3/5 | 1.4 |
| Standard Dev | 1.297574545 | 0.9772916402 | 0.5439900734 | 1.122309449 | 1.990699299 | 0.615091461 | 6.323076869 | | | 0.5477225575 |

Triangle Prism

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 9.48 | 8.812333333 | 3.505666667 | 29.738 | 33.963 | 3.136919831 | 87.55999176 | yes | no | 2 |
| 2 | 4.172 | 5.824 | 4.313333333 | 17.226 | 34.775 | 4.128954938 | 49.53558591 | yes | YES | 1 |
| 3 | 4.712666667 | 5.212333333 | 4.742666667 | 32.175 | 35.75 | 6.827344745 | 90 | yes | yes | 1 |
| 4 | 5.579333333 | 9.340666667 | 3.125 | 20.313 | 34.125 | 3.640757558 | 59.52527473 | yes | no | 4 |
| 5 | 8.179333333 | 7.060666667 | 3.091333333 | 21.938 | 35.1 | 2.682125683 | 62.5014245 | yes | no | 1 |
| AVG= | 6.424666667 | 7.25 | 3.7556 | 24.278 | 34.7426 | 4.083220551 | 69.82445538 | 5/5 | 2/5 | 1.8 |
| Standard Dev | 2.298584154 | 1.805074868 | 0.7393815509 | 6.38556454 | 0.730213188 | 1.626855449 | 17.9787373 | | | 1.303840481 |

Square Pyramid

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.991333333 | 4.456333333 | 2.494 | 24.863 | 30.875 | 3.556260132 | 80.52793522 | No | No | 3 |
| 2 | 7.779333333 | 7.044 | 2.109333333 | 23.077 | 32.313 | 2.966449567 | 71.41707672 | No | No | 2 |
| 3 | 5.788333333 | 7.628666667 | 2.421 | 21.135 | 31.204 | 3.651310107 | 67.73170106 | Yes | Yes | 4 |
| 4 | 7.535333333 | 4.034 | 1.459333333 | 11.05 | 32.988 | 1.466424843 | 33.49702922 | Yes | Yes | 2 |
| 5 | 6.068 | 5.476666667 | 1.924666667 | 22.1 | 29.575 | 3.642056691 | 74.72527473 | yes | Yes | 2 |
| AVG= | 6.832466667 | 5.727933333 | 2.081666667 | 20.445 | 31.391 | 3.056500268 | 65.57980339 | 3/5 | 3/5 | 2.6 |
| Standard Dev | 0.8789824167 | 1.572762721 | 0.4176957957 | 5.429716797 | 1.322835402 | 0.9331243186 | 18.54113746 | | | 0.894427191 |

| 2:1 | HemiEllipsoids | cone | Hyperbolic Cone | Triangular Prism | Square Pyramid |
|---|---|---|---|---|---|
| Base AVG | 6.934733333 | 7.031933333 | 6.3712 | 6.424666667 | 6.832466667 |
| Base DEV | 0.3773597164 | 0.4423592934 | 1.297574545 | 2.298584154 | 0.8789824167 |
| Transition AVG | 6.6307 | 6.123266667 | 6.4403 | 7.25 | 5.727933333 |
| Transition DEV | 0.8631107853 | 0.4527054721 | 0.9772916402 | 1.805074868 | 1.572762721 |
| Tip AVG | 3.330466667 | 1.51987 | 1.3363 | 3.7556 | 2.081666667 |
| Tip Dev | 1.10289177 | 0.4051710267 | 0.5439900734 | 0.7393815509 | 0.4176957957 |
| Nuc/base AVG | 4.911722859 | 2.358411004 | 1.671131595 | 4.083220551 | 3.056500268 |
| Nuc/base DEV | 0.5842724888 | 0.5842724888 | 0.615091461 | 1.626855449 | 0.9331243186 |
| Nuc/height % | 93.1687807 | 48.93946655 | 38.97864309 | 69.82445538 | 65.57980339 |
| Nuc/height dev | 6.757079095 | 12.01128079 | 6.323076869 | 17.9787373 | 18.54113746 |
| Average # of nuclei | 4.4 | 2.2 | 1.4 | 1.8 | 2.6 |
| change in thickn. | 3.604266667 | 5.512063333 | 5.0349 | 2.669066667 | 4.7508 |

1:1

HemiEllipsoids

| | Base | Transition | Tip | nucleus | height | nuc/base | nuc/height | reach at top | over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 9.711 | 5.961 | 2.284 | 9.75 | 20.8 | 1.004016064 | 46.875 | yes | yes | 3 |
| 2 | 8.762 | 10.811 | 1.979 | 14.625 | 18.525 | 1.669139466 | 78.94736842 | yes | yes | 1 |
| 3 | 13.341 | 5.72 | 2.6 | 14.625 | 18.85 | 1.096244659 | 77.5862069 | yes | yes | 1 |
| 4 | 6.673 | 7.942 | 3.004 | 16.25 | 19.5 | 2.435186573 | 83.33333333 | yes | yes | 2 |
| 5 | 3.996 | 5.6025 | 1.6665 | 13.975 | 16.25 | 3.497247247 | 86 | yes | yes | 2 |
| AVG | 8.4966 | 7.2073 | 2.3067 | 13.845 | 18.785 | 1.940366802 | 74.54838173 | 5/5 | 5/5 | 1.8 |
| standard dev | 3.485481789 | 2.228628827 | 0.5220179116 | 2.438583093 | 1.663542906 | 1.040622709 | 15.83280185 | | | 0.8366600265 |

Cone

| | Base | Transition | Tip | nucleus | height | nuc/base | Nuc/Height % | Reach at top | over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.8 | 7.687 | 2.7145 | 10.725 | 16.9 | 1.375 | 63.46153846 | yes | no | 2 |
| 2 | 6.175 | 11.183 | 3.428 | 12.675 | 19.5 | 2.052631579 | 65 | yes | no | 3 |
| 3 | 5.525 | 3.852 | 1.75 | 10.563 | 16.088 | 1.911855204 | 65.65763302 | yes | yes | 1 |
| 4 | 7.475 | 4.495 | 1.09 | 11.538 | 16.413 | 1.543545151 | 70.29793456 | yes | yes | 1 |
| 5 | 7.475 | 5.45 | 2.035 | 14.138 | 16.088 | 1.891371237 | 87.87916459 | yes | yes | 1 |
| AVG | 6.89 | 6.5334 | 2.2035 | 11.9278 | 16.9978 | 1.754880634 | 70.45925413 | 5/5 | 3/5 | 1.6 |
| standard dev | 0.9857738077 | 2.977763809 | 0.8992299483 | 1.492032406 | 1.437709359 | 0.2832394342 | 10.06611084 | | | 0.894427191 |

Hyperbolics

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.9 | 3.6885 | 1.0855 | 4.388 | 24.96 | 0.7437288136 | 17.58012821 | yes | no | 1 |
| 2 | 7.797 | 10.341 | 1.814 | 13.488 | 14.465 | 1.729896114 | 93.24576564 | yes | yes | 1 |
| 3 | 4.954 | 5.1465 | 0.403 | 5.688 | 14.4655 | 1.148163101 | 39.32114341 | yes | no | 0 |
| 4 | 13.813 | 9.013 | 1.065 | 14.138 | 14.463 | 1.02352856 | 97.75288668 | yes | yes | 2 |
| 5 | 10.15 | 7.312 | 2.868666667 | 11.05 | 17.55 | 1.088669951 | 62.96296296 | yes | yes | 3 |
| AVG= | 8.5228 | 7.1002 | 1.447233333 | 9.7504 | 17.1807 | 1.146797308 | 62.17257738 | 5/5 | 3/5 | 1.4 |
| Standard Dev | 3.562870851 | 2.723191524 | 0.9383985146 | 4.476831335 | 4.549371435 | 0.3609169807 | 34.43399739 | | | 1.140175425 |

Triangle Prism

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.22 | 7.4875 | 3.901 | 19.013 | 19.5 | 2.633379501 | 97.5025641 | yes | no | 1 |
| 2 | 9.627 | 5.755 | 2.17 | 18.038 | 21.54 | 1.873688584 | 83.74187558 | yes | no | 2 |
| 3 | 5.075 | 8.1495 | 1.947 | 16.738 | 22.1 | 3.298128079 | 75.73755656 | yes | yes | 2 |
| 4 | 5.291 | 9.0835 | 2.72 | 15.925 | 21.45 | 3.00982801 | 74.24242424 | yes | yes | 3 |
| 5 | 7.676 | 6.2585 | 2.499 | 28.15 | 22.05 | 3.667274622 | 127.6643991 | yes | yes | 2 |
| AVG= | 6.9778 | 7.3468 | 2.6474 | 19.5728 | 21.328 | 2.896459759 | 91.77776392 | 5/5 | 3/5 | 2 |
| Standard Dev | 1.872855227 | 1.360006875 | 0.7611302779 | 4.939126512 | 1.062859351 | 0.6862247173 | 22.07926523 | | | 0.7071067812 |

Square Pyramid

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.349 | 6.193 | 1.389 | 5.038 | 17.713 | 0.685535447 | 28.44238695 | yes | no | 0 |
| 2 | 6.165 | 5.884 | 1.761 | 5.04 | 18.2 | 0.8175182482 | 27.69230769 | yes | no | 0 |
| 3 | 6.392 | 6.0995 | 1.205 | 10.075 | 17.55 | 1.576188986 | 57.40740741 | yes | no | 1 |
| 4 | 9.3835 | 7.656 | 1.69 | 10.888 | 16.9 | 1.16033463 | 64.4260355 | yes | no | 0 |
| 5 | 9.601 | 7.6615 | 1.829 | 5.85 | 16.413 | 0.60931153 | 35.64247852 | yes | no | 1 |
| AVG= | 7.7781 | 6.6988 | 1.5748 | 7.3782 | 17.3552 | 0.9697777683 | 42.72212321 | 5/5 | 0/5 | 0.4 |
| Standard Dev | 1.628478216 | 0.8834472678 | 0.266379804 | 2.866641624 | 0.7026028039 | 0.3993616939 | 17.07809249 | | | 0.5477225575 |

| 1:1 | HemiEllipsoids | cone | Hyperbolic Cone | Triangular Prism | Square Pyramid |
|---|---|---|---|---|---|
| Base AVG | 8.4966 | 6.89 | 8.5228 | 6.9778 | 7.7781 |
| Base DEV | 3.485481789 | 0.9857738077 | 3.562870851 | 1.872855227 | 1.628478216 |
| Transition AVG | 7.2073 | 6.5334 | 7.1002 | 7.3468 | 6.6988 |
| Transition DEV | 2.228628827 | 2.977763809 | 2.723191524 | 1.360006875 | 0.8834472678 |
| Tip AVG | 2.3067 | 2.2035 | 1.447233333 | 2.6474 | 1.5748 |
| Tip Dev | 0.5220179116 | 0.8992299483 | 0.9383985146 | 0.7611302779 | 0.266379804 |
| Nuc/base AVG | 1.754880634 | 1.754880634 | 1.146797308 | 2.896459759 | 0.9697777683 |
| Nuc/base DEV | 1.040622709 | 0.2832394342 | 0.3609169807 | 0.6862247173 | 0.3993616939 |
| Nuc/height % | 74.54838173 | 70.45925413 | 62.17257738 | 91.77776392 | 42.72212321 |
| Nuc/height dev | 15.83280185 | 10.06611084 | 34.43399739 | 22.07926523 | 17.07809249 |
| Average # of nuclei | 1.8 | 1.6 | 1.4 | 2 | 0.4 |
| change in thick. | 6.1899 | 4.6865 | 7.075566667 | 4.3304 | 6.2033 |

2:1

HemiEllipsoids

| | Base | Transition | Tip | nucleus | height | nuc/base | nuc/height | reach at top | over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 3.9 | 5.579 | 1.654 | 31.525 | 34.883 | 8.083333333 | 90.37353439 | yes | yes | 4 |
| 2 | 4.333 | 6.1515 | 1.473 | 26.325 | 34.884 | 6.075467344 | 75.46439628 | yes | yes | 4 |
| 3 | 1.95 | 6.3985 | 2.645 | 33.908 | 33.8 | 17.38871795 | 100.3195266 | yes | yes | 3 |
| 4 | 2.2 | 5.637 | 2.303 | 30.442 | 32.283 | 13.83727273 | 94.29730818 | yes | yes | 8 |
| 5 | 3.667 | 5.3505 | 1.938 | 28.058 | 33.8 | 7.651486229 | 83.01183432 | yes | yes | 4 |
| AVG | 3.21 | 5.8233 | 2.0026 | 30.0516 | 33.93 | 10.60725552 | 88.69331996 | 5/5 | 5/5 | 4.6 |
| standard dev | 1.066979147 | 0.4348851285 | 0.4767308465 | 2.960453057 | 1.068261906 | 4.797528093 | 9.704125239 | | | 1.949358869 |

Cone

| | Base | Transition | Tip | nucleus | height | nuc/base | Nuc/Height % | Reach at top | over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.2 | 6.569666667 | 1.5795 | 15.167 | 29.142 | 2.916730769 | 52.04515819 | yes | no | 3 |
| 2 | 7.367 | 5.04 | 1.355 | 21.233 | 30.875 | 2.882177277 | 68.7708502 | yes | no | 3 |
| 3 | 4.983 | 4.006 | 2.373 | 15.167 | 31.417 | 3.043748746 | 48.27641086 | yes | no | 3 |
| 4 | 3.863 | 5.936 | 1.602 | 17.767 | 30.333 | 4.599275175 | 58.57317113 | yes | no | 3 |
| 5 | 7.357 | 4.896 | 1.3315 | 12.783 | 29.863 | 1.737528884 | 42.80547835 | yes | no | 3 |
| AVG | 5.754 | 5.289533333 | 1.6482 | 16.4234 | 30.326 | 3.03589217 | 54.09421375 | 5/5 | 0/5 | 3 |
| standard dev | 1.553111071 | 0.9901077831 | 0.4238084178 | 3.215085815 | 0.8814697953 | 1.020759898 | 10.01181456 | | | 0 |

Hyperbolics

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 8.017 | 4.946 | 1.2265 | 10.617 | 23.184 | 1.324310839 | 45.79451346 | yes | no | 2 |
| 2 | 5.85 | 6.435 | 0.9665 | 8.45 | 25.567 | 1.444444444 | 33.05041655 | yes | yes | 2 |
| 3 | 4.983 | 5.133 | 1.089 | 5.421 | 24.483 | 1.087898856 | 22.14189438 | yes | no | 3 |
| 4 | 6.067 | 3.8675 | 0.789 | 6.5 | 24.917 | 1.071369705 | 26.08660754 | yes | yes | 2 |
| 5 | 7.583 | 6.321 | 1.082 | 9.967 | 19.5 | 1.314387446 | 51.11282051 | yes | no | 3 |
| AVG= | 6.5 | 5.3405 | 1.0306 | 8.191 | 23.5302 | 1.248482258 | 35.63725049 | 5/5 | 2/5 | 2.4 |
| Standard Dev | 1.263472991 | 1.063922695 | 0.1634844794 | 2.215801999 | 2.415676034 | 0.1625196384 | 12.47692256 | | | 0.5477225575 |

Triangle Prism

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.392 | 6.312 | 3.1765 | 19.069 | 32.283 | 2.983260325 | 59.06824025 | yes | no | 3 |
| 2 | 5.308 | 8.7155 | 2.038 | 21.775 | 31.2 | 4.102298417 | 69.79166667 | yes | no | 1 |
| 3 | 3.575 | 7.2025 | 2.513 | 24.483 | 30.875 | 6.848391608 | 79.29716599 | yes | yes | 3 |
| 4 | 6.392 | 6.368 | 2.88885 | 24.592 | 30.225 | 3.847309136 | 81.36311001 | yes | yes | 3 |
| 5 | 5.742 | 9.2675 | 3.6685 | 29.25 | 32.392 | 5.094043887 | 90.30007409 | yes | no | 2 |
| AVG= | 5.4818 | 7.5731 | 2.85697 | 23.8338 | 31.395 | 4.575060675 | 75.9640514 | 5/5 | 2/5 | 2.4 |
| Standard Dev | 1.160896722 | 1.356102199 | 0.6226403292 | 3.783963755 | 0.9300373648 | 1.476849003 | 11.93050503 | | | 0.894427191 |

Square Pyramid

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.633 | 5.8085 | 2.269 | 19.938 | 29.901 | 3.539499379 | 66.68004415 | yes | no | 3 |
| 2 | 5.417 | 6.021 | 1.419 | 14.733 | 29.9 | 2.719771091 | 49.27424749 | yes | no | 3 |
| 3 | 3.9 | 5.825 | 1.639 | 14.3 | 30.393 | 3.666666667 | 47.05030764 | yes | no | 2 |
| 4 | 12.142 | 4.575 | 0.893 | 15.167 | 30.938 | 1.249135233 | 49.02385416 | yes | no | 1 |
| 5 | 7.8 | 5.9805 | 1.02 | 15.387 | 31.666 | 1.972692308 | 48.5915493 | yes | no | 1 |
| AVG= | 6.9784 | 5.642 | 1.448 | 15.905 | 30.5596 | 2.629552935 | 52.12400055 | 5/5 | 0/5 | 2 |
| Standard Dev | 3.203997862 | 0.6037225149 | 0.5483092193 | 2.292796655 | 0.7523113052 | 1.030626709 | 8.182771183 | | | 1 |

| 2:1 | HemiEllipsoids | cone | Hyperbolic Cone | Triangular Prism | Square Pyramid |
|---|---|---|---|---|---|
| Base AVG | 3.21 | 5.754 | 6.5 | 5.4818 | 6.9784 |
| Base DEV | 1.066979147 | 1.553111071 | 1.263472991 | 1.160896722 | 3.203997862 |
| Transition AVG | 5.8233 | 5.289533333 | 5.3405 | 7.5731 | 5.642 |
| Transition DEV | 0.4348851285 | 0.9901077831 | 1.063922695 | 1.356102199 | 0.6037225149 |
| Tip AVG | 2.0026 | 1.6482 | 1.0306 | 2.85697 | 1.448 |
| Tip Dev | 0.4767308465 | 0.4238084178 | 0.1634844794 | 0.6226403292 | 0.5483092193 |
| Nuc/base AVG | 10.60725552 | 3.03589217 | 1.248482258 | 4.575060675 | 2.629552935 |
| Nuc/base DEV | 1.020759898 | 1.020759898 | 0.1625196384 | 1.476849003 | 1.030626709 |
| Nuc/height % | 88.69331996 | 54.09421375 | 35.63725049 | 75.9640514 | 52.12400055 |
| Nuc/height dev | 9.704125239 | 10.01181456 | 12.47692256 | 11.93050503 | 8.182771183 |
| Average # of nuclei | 4.6 | 3 | 2.4 | 2.4 | 2 |
| Change in thickness | 1.2074 | 4.1058 | 5.4694 | 2.62483 | 5.5304 |

1:1

HemiEllipsoids

| | Base | Transition | Tip | nucleus | height | nuc/base | nuc/height | reach at top | over? | # of nuclei on structure | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 7.583 | 2.3835 | 0.7585 | 7.042 | 17.225 | 0.9286562047 | 40.88243832 | yes | yes | 2 | |
| 2 | 6.933 | 3.9 | 1.4625 | 18.308 | 15.492 | 2.64070388 | 118.1771237 | yes | yes | 5 | has a nucleus on top! |
| 3 | 6.067 | 2.3835 | 1.5165 | 9.208 | 16.358 | 1.517718807 | 56.29050006 | yes | yes | 3 | |
| 4 | 7.15 | 1.0835 | 0.758 | 10.292 | 17.658 | 1.439440559 | 58.28519651 | yes | yes | 1 | |
| 5 | 6.5 | 2.0585 | 0.8125 | 12.675 | 16.792 | 1.95 | 75.48237256 | yes | yes | 2 | |
| AVG | 6.8466 | 2.3618 | 1.0616 | 11.505 | 16.705 | 1.69530389 | 69.82352622 | yes | yes | 2.6 | |
| standard dev | 0.5851728804 | 1.011563493 | 0.391710416 | 3.855273168 | 0.8333600662 | 0.6409762922 | 29.68145611 | 5/5 | 5/5 | 1.516575089 | |

Cone

| | Base | Transition | Tip | nucleus | height | nuc/base | Nuc/Height % | Reach at top | over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.717 | 7.5695 | 3.4015 | 10.4 | 14.3 | 1.548310258 | 72.72727273 | yes | yes | 2 |
| 2 | 7.15 | 6.743 | 2.716 | 4.983 | 14.3 | 0.6969230769 | 34.84615385 | yes | yes | 2 |
| 3 | 9.75 | 9.0135 | 2.1055 | 5.85 | 14.733 | 0.6 | 39.7067807 | yes | yes | 1 |
| 4 | 6.067 | 5.056 | 1.962 | 3.25 | 14.95 | 0.5356848525 | 21.73913043 | yes | no | 2 |
| 5 | 5.2 | 9.6285 | 3.1615 | 6.504 | 15.383 | 1.250769231 | 42.28043945 | yes | no | 1 |
| AVG | 6.9768 | 7.6021 | 2.6693 | 6.1974 | 14.7332 | 0.9263374835 | 42.25995543 | 5/5 | 3/5 | 1.6 |
| standard dev | 1.71558203 | 1.824834766 | 0.6321989995 | 2.647568281 | 0.4595015778 | 0.4482863737 | 18.78061006 | | | 0.5477225575 |

Hyperbolics

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 5.633 | 4.7465 | 1.332 | 2.85 | 13 | 0.5059470975 | 21.92307692 | yes | no | 0 |
| 2 | 8.883 | 9.9955 | 3.113 | 9.1 | 13.217 | 1.024428684 | 68.85072255 | yes | yes | 3 |
| 3 | 6.933 | 5.4815 | 0.807 | 5.421 | 13 | 0.781912592 | 41.7 | yes | yes | 2 |
| 4 | 7.853 | 5.6375 | 1.1305 | 3.494 | 13.217 | 0.4449255062 | 26.43565106 | yes | yes | 0 |
| 5 | 6.5 | 4.316 | 1.131 | 4.55 | 13.217 | 0.7 | 34.42536128 | yes | yes | 0 |
| AVG= | 7.1604 | 6.0354 | 1.5027 | 5.083 | 13.1302 | 0.6914427759 | 38.66696236 | 5/5 | 4/5 | 1 |
| Standard Dev | 1.251782249 | 2.278438127 | 0.919632644 | 2.451827482 | 0.118855795 | 0.231471942 | 18.49467864 | | | 1.414213562 |

Triangle Prism

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 6.224 | 7.1585 | 1.064 | 13.975 | 17.658 | 2.245340617 | 79.14259826 | yes | no | 1 |
| 2 | 7.042 | 6.148 | 2.1615 | 7.583 | 16.25 | 1.076824766 | 46.66461538 | yes | yes | 2 |
| 3 | 5.417 | 6.1045 | 1.324 | 5.633 | 16.575 | 1.039874469 | 33.98491704 | yes | yes | 0 |
| 4 | 7.042 | 6.184 | 2.968 | 8.342 | 17.225 | 1.184606646 | 48.42960813 | yes | yes | 1 |
| 5 | 7.692 | 6.836 | 2.8605 | 14.625 | 17.658 | 1.901326053 | 82.82364934 | yes | yes | 2 |
| AVG= | 6.6834 | 6.4862 | 2.0756 | 10.0316 | 17.0732 | 1.48959451 | 58.20907763 | 5/5 | 4/5 | 1.2 |
| Standard Dev | 0.8788172734 | 0.4810785539 | 0.8671882293 | 4.026406562 | 0.6389222957 | 0.549162912 | 21.56276892 | | | 0.8366600265 |

Square Pyramid

| | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? | # of nuclei on structure |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 4.55 | 6.481 | 1.394 | 3.69 | 15.6 | 0.810989011 | 23.65384615 | yes | no | 0 |
| 2 | 5.85 | 3.951 | 1.332 | 7.367 | 15.383 | 1.259316239 | 47.89052851 | yes | yes | 0 |
| 3 | 3.25 | 6.0995 | 1.9415 | 5.85 | 15.167 | 1.8 | 38.57058087 | yes | no | 2 |
| 4 | 9.967 | 5.4425 | 1.682 | 9.533 | 14.3 | 0.9564563058 | 66.66433566 | yes | no | 2 |
| 5 | 6.717 | 4.0955 | 2.5025 | 9.967 | 14.95 | 1.483846955 | 66.66889632 | yes | no | 2 |
| AVG= | 6.0668 | 5.2139 | 1.7704 | 7.2814 | 15.08 | 1.262121702 | 48.6896375 | 5/5 | 1/5 | 1.2 |
| Standard Dev | 2.545405999 | 1.149763313 | 0.4761648087 | 2.609431796 | 0.4987679821 | 0.3985709038 | 18.5483709 | | | 1.095445115 |

| 1:1 | HemiEllipsoids | cone | Hyperbolic Cone | Triangular Prism | Square Pyramid |
|---|---|---|---|---|---|
| Base AVG | 6.8466 | 6.9768 | 7.1604 | 6.6834 | 6.0668 |
| Base DEV | 0.5851728804 | 1.71558203 | 1.251782249 | 0.8788172734 | 2.545405999 |
| Transition AVG | 2.3618 | 7.6021 | 6.0354 | 6.4862 | 5.2139 |
| Transition DEV | 1.011563493 | 1.824834766 | 2.278438127 | 0.4810785539 | 1.149763313 |
| Tip AVG | 1.0616 | 2.6693 | 1.5027 | 2.0756 | 1.7704 |
| Tip Dev | 0.391710416 | 0.6321989995 | 0.919632644 | 0.8671882293 | 0.4761648087 |
| Nuc/base AVG | 0.9263374835 | 0.9263374835 | 0.6914427759 | 1.48959451 | 1.262121702 |
| Nuc/base DEV | 0.6409762922 | 0.4482863737 | 0.231471942 | 0.549162912 | 0.3985709038 |
| Nuc/height % | 69.82352622 | 42.25995543 | 38.66696236 | 58.20907763 | 48.6896375 |
| Nuc/height dev | 29.68145611 | 18.78061006 | 18.49467864 | 21.56276892 | 18.5483709 |
| Average # of nuclei | 2.6 | 1.6 | 1 | 1.2 | 1.2 |
| change in thick. | 5.785 | 4.3075 | 5.6577 | 4.6078 | 4.2964 |

| | | | | | |
|---|---|---|---|---|---|
| HemiEllipsoids | Width | Height | | True Height | 15 |
| 1 | 15.06 | 15.999 | | True Width: | 15 |
| 2 | 14.343 | 16.999 | | | |
| 3 | 15.06 | 16.238 | | | |
| 4 | 14.343 | 13.999 | | | |
| | 14.343 | 19.363 | | | |
| | 15.06 | 17.928 | | | |
| | 14.353 | 17.549 | | | |
| | 14.343 | 17.928 | | | |
| | 15.06 | 18.646 | | | |
| 4 | 13.626 | 17.23 | | | |
| Average | 14.5591 | 17.1879 | | | |
| STDEV | 0.4834543412 | 1.513103393 | | | |
| | | | | | |
| Cone | | | | True Height | 15 |
| 1 | 12.908 | 12.999 | | True Width: | 15 |
| 1 | 12.191 | 12.999 | | | |
| 2 | 13.626 | 13.999 | | | |
| 1 | 13.626 | 12.951 | | | |
| | 12.908 | 13.626 | | | |
| | 13.626 | 12.908 | | | |
| | 11.474 | 13.626 | | | |
| | 13.626 | 13.077 | | | |
| | 14.343 | 13.626 | | | |
| 2 | 11.514 | 13.999 | | | |
| Average | 12.9842 | 13.381 | | | |
| STDEV | 0.9762136378 | 0.4392943837 | | | |
| | | | | | |
| Hyperbolic | | | | True Height | 15 |
| 1 | 12.191 | 10.999 | | True Width: | 15 |
| 2 | 12.191 | 11.231 | | | |
| 1 | 12.191 | 10.999 | | | |

| | | | | | |
|---|---|---|---|---|---|
| 1 | 12.908 | 10.999 | | | |
| | 12.908 | 10.575 | | | |
| | 13.626 | 10.757 | | | |
| | 11.474 | 11.474 | | | |
| | 12.908 | 11.474 | | | |
| | 13.626 | 11.474 | | | |
| 1 | 10.757 | 10.999 | | | |
| Average | 12.478 | 11.0981 | | | |
| STDEV | 0.9072223787 | 0.3110610694 | | | |
| | | | | | |
| Prism | | | | True Height | 15 |
| 1 | 12.191 | 15.999 | | True Width: | 15 |
| 1 | 12.908 | 15.999 | | | |
| 1 | 12.908 | 14.999 | | | |
| 1 | 12.908 | 14.999 | | | |
| | 12.908 | 15.06 | | | |
| | 13.626 | 15.06 | | | |
| | 12.908 | 15.777 | | | |
| | 13.626 | 15.777 | | | |
| | 12.908 | 15.06 | | | |
| 1 | 12.191 | 14.999 | | | |
| Average | 12.9082 | 15.3729 | | | |
| STDEV | 0.478333403 | 0.4501504563 | | | |
| | | | | | |
| Pyramid | | | | | |
| 1 | 14.343 | 14.999 | | True Height | 15 |
| 1 | 13.626 | 15.999 | | True Width: | 15 |
| 2 | 13.626 | 15.999 | | | |
| 1 | 14.343 | 15.999 | | | |
| | 13.626 | 15.777 | | | |
| | 14.343 | 15.777 | | | |

| | | | | | |
|---|---|---|---|---|---|
| | 12.908 | 16.494 | | | |
| | 14.343 | 16.494 | | | |
| | 14.343 | 15.06 | | | |
| 1 | 12.908 | 16.999 | | | |
| Average | 13.8409 | 15.9597 | | | |
| STDEV | 0.5906374052 | 0.6195781809 | | | |

| | | | | | |
|---|---|---|---|---|---|
| HemiEllipsoids | | | | True Height | 30 |
| | Width | Height | | True Width: | 30 |
| 1 | 31.554 | 27.998 | | | |
| 2 | 31.554 | 28.998 | | | |
| 3 | 30.837 | 27.998 | | | |
| 4 | 29.403 | 27.878 | | | |
| 5 | 29.403 | 29.998 | | | |
| 6 | 28.523 | 27.838 | | | |
| 7 | 29.061 | 26.748 | | | |
| 8 | 28.523 | 30.998 | | | |
| | 29.403 | 28.685 | | | |
| 9 | 30.675 | 27.998 | | | |
| Average | 29.8936 | 28.5137 | | | |
| STDEV | 1.163654321 | 1.220883291 | | | |
| | | | | | |
| Cone | | | | True Height | 30 |
| 1 | 15.777 | 25.998 | | True Width: | 15 |
| 2 | 14.343 | 24.998 | | | |
| 3 | 15.06 | 24.998 | | | |
| 4 | 15.777 | 25.998 | | | |
| 5 | 14.343 | 25.817 | | | |
| 6 | 15.06 | 26.534 | | | |
| 7 | 15.06 | 26.534 | | | |
| 8 | 13.626 | 17.251 | | | |
| 9 | 15.777 | 26.534 | | | |
| 10 | 15.777 | 26.998 | | | |
| Average | 15.06 | 25.166 | | | |
| STDEV | 0.7557843608 | 2.857396212 | | | |
| | | | | | |
| Hyperbolics | | | | True Height | 30 |
| 1 | 12.908 | 22.998 | | True Width: | 15 |
| 2 | 12.191 | 22.998 | | | |

| | | | | | |
|---|---|---|---|---|---|
| 3 | 12.191 | 22.998 | | | |
| 4 | 11.474 | 23.998 | | | |
| 5 | 12.191 | 22.948 | | | |
| 6 | 12.908 | 23.665 | | | |
| 7 | 12.908 | 23.665 | | | |
| 8 | 14.343 | 24.383 | | | |
| 9 | 15.777 | 25.817 | | | |
| 10 | 14.343 | 22.998 | | | |
| Average | 13.1234 | 23.6468 | | | |
| STDEV | 1.311668251 | 0.9157783332 | | | |
| | | | | | |
| Triangular Prism | | | | True Height | 30 |
| 1 | 16.494 | 28.998 | | True Width: | 15 |
| 2 | 17.211 | 27.998 | | | |
| 3 | 13.626 | 27.998 | | | |
| 4 | 12.908 | 29.998 | | | |
| | 17.211 | 27.968 | | | |
| | 19.363 | 27.251 | | | |
| | 20.797 | 27.968 | | | |
| | 17.928 | 27.251 | | | |
| | 15.06 | 25.817 | | | |
| 5 | 13.626 | 28.998 | | | |
| Average | 16.4224 | 28.0245 | | | |
| STDEV | 2.60654813 | 1.148310667 | | | |
| | | | | | |
| Square Pyramid | | | | True Height | 30 |
| 1 | 17.291 | 27.998 | | True Width | 15 |
| 2 | 18.646 | 27.998 | | | |
| 3 | 15.777 | 26.998 | | | |
| 4 | 15.777 | 25.998 | | | |
| | 14.343 | 25.817 | | | |

| | | | | | |
|---|---|---|---|---|---|
| | 15.06 | 26.534 | | | |
| | 14.343 | 25.1 | | | |
| | 15.06 | 26.534 | | | |
| | 12.908 | 25.1 | | | |
| 5 | 12.908 | 26.998 | | | |
| Average | 15.2113 | 26.5075 | | | |
| STDEV | 1.792775753 | 0.9810330525 | | | |

| | | | | | |
|---|---|---|---|---|---|
| HemiEllipsoids | | | | True Height | 30 |
| | Width | Height | | True Width: | 30 |
| 1 | 29.252 | 34.125 | | | |
| 2 | 27.95 | 34.938 | | | |
| 3 | 27.463 | 33.638 | | | |
| 4 | 28.113 | 33.8 | | | |
| 5 | 28.6 | 33.962 | | | |
| Average | | | | | |
| STDEV | 0.6803244079 | 0.5062319626 | | | |
| | | | | | |
| Cone | | | | True Height | 30 |
| | Width | Height | | True Width: | 15 |
| 1 | 15.925 | 28.762 | | | |
| 2 | 12.513 | 37.7 | | | |
| 3 | 13 | 31.525 | | | |
| 4 | 14.625 | 29.25 | | | |
| 5 | 13.975 | 32.175 | | | |
| Average | | | | | |
| STDEV | 1.351643 | 3.561047781 | | | |
| | | | | | |
| | | | | | |
| Hyperbolics | | | | True Height | 30 |
| | Width | Height | | True Width: | 15 |
| 1 | 8.45 | 26.195 | | | |
| 2 | 8.613 | 28.6 | | | |
| 3 | 8.125 | 28.275 | | | |
| 4 | 8.613 | 24.7 | | | |
| 5 | 13 | 22.75 | | | |
| Average | | | | | |
| STDEV | 2.044440192 | 2.458254971 | | | |
| | | | | | |
| | | | | | |

| | | | | | |
|---|---|---|---|---|---|
| Triangular Prisms | | | | True Height | 30 |
| | Width | Height | | True Width: | 15 |
| 1 | 13.375 | 31.2 | | | |
| 2 | 11.375 | 33.312 | | | |
| 3 | 11.537 | 32.825 | | | |
| 4 | 12.351 | 34.613 | | | |
| 5 | 12.35 | 34.938 | | | |
| Average | | | | | |
| STDEV | 0.7978237901 | 1.501055728 | | | |
| | | | | | |
| | | | | | |
| | | | | | |
| Square Pyramid | | | | True Height | 30 |
| | Width | Height | | True Width: | 15 |
| 1 | 13 | 28.925 | | | |
| 2 | 13.488 | 30.387 | | | |
| 3 | 13.325 | 30.387 | | | |
| 4 | 14.3 | 32.988 | | | |
| 5 | 14.3 | 33.962 | | | |
| Average | 13.6826 | | | | |
| STDEV | 0.5903505738 | 2.075988367 | | | |

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| HemiEllipsoids | | | | True Height | 15 | | HemiEllipsoids | | | | | | | | |
| | Width | Height | | True Width: | 15 | | | Base | Transition | Tip | | | | | |
| 1 | 13.65 | 18.2 | | | | | 1 | 6.988 | | | | | | | |
| 2 | 14.3 | 20.312 | | | | | 2 | | | | | | | | |
| 3 | 15.113 | 20.8 | | | | | 3 | | | | | | | | |
| 4 | 14.3 | 19.663 | | | | | 4 | | | | | | | | |
| 5 | 13.975 | 19.175 | | | | | 5 | | | | | | | | |
| Average | | | | | | | | | | | | | | | |
| STDEV | 0.5440232532 | 1.010821201 | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| Cone | | | | True Height | 15 | | Cone | | | | | | | | |
| | Width | Height | | True Width: | 15 | | | Base | Transition | Tip | nucleus | height | nuc/base | | |
| 1 | 12.514 | 19.175 | | | | | 1 | 6.991 | 6.854 | 1.15 | | | | | |
| 2 | 14.787 | 20.15 | | | | | 2 | | | | | | | | |
| 3 | 12.513 | 20.8 | | | | | 3 | | | | | | | | |
| 4 | 15.6 | 20.8 | | | | | 4 | | | | | | | | |
| 5 | 14.787 | 16.575 | | | | | 5 | | | | | | | | |
| Average | | | | | | | | | | | | | | | |
| STDEV | 1.432656868 | 1.76520183 | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| Hyperbolics | | | | True Height | 15 | | Hyperbolics | | | | | | | | |
| | Width | Height | | True Width: | 15 | | | Base | Transition | Tip | Nucleus | Height | Nuc/base | Nuc/Height % | Reach at top | Over? |
| 1 | 10.4 | 18.038 | | | | | 1 | 7.731 | 5.677 | 0.998 | 10.239 | 24.7 | 1.324408227 | 41.4534413 | No | No |
| 2 | 11.7 | 16.738 | | | | | 2 | 6.554 | 7.0135 | 1.425 | 8.775 | 28.275 | 1.338877022 | 31.03448276 | No | No |
| 3 | 13.162 | 18.85 | | | | | 3 | 6.988 | 7.509 | 0.711 | 9.263 | 27.625 | 1.3255581 | 33.53122172 | Yes | Yes |
| 4 | 11.537 | 18.525 | | | | | 4 | 4.257 | 5.17 | 2.153 | 11.7 | 25.838 | 2.748414376 | 45.28214258 | Yes | Yes |
| 5 | 10.4 | 16.738 | | | | | 5 | 6.326 | 6.832 | 1.3945 | 10.238 | 23.486 | 1.618400253 | 43.59192711 | yes | Yes |
| Average | | | | | | | AVG= | 6.3712 | 6.4403 | 1.3363 | 10.043 | 25.9848 | 1.671131595 | 38.97864309 | | |
| STDEV | 1.140777454 | 0.9922188267 | | | | | Standard Dev | 1.297574545 | 0.9772916402 | 0.5439900734 | 1.122309449 | 1.990699299 | 0.615091461 | 6.323076869 | | |
| | | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | | |
| Triangular Prisms | | | | True Height | 15 | | | | | | | | | | |
| | Width | Height | | True Width: | 15 | | | | | | | | | | |
| 1 | 11.537 | 17.713 | | | | | | | | | | | | | |
| 2 | 14.138 | 20.312 | | | | | | | | | | | | | |
| 3 | 12.35 | 21.288 | | | | | | | | | | | | | |
| 4 | 15.113 | 22.263 | | | | | | | | | | | | | |
| 5 | 11.05 | 21.94 | | | | | | | | | | | | | |
| Average | | | | | | | | | | | | | | | |
| STDEV | 1.731568162 | 1.830230505 | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| | | | | | | | | | | | | | | | |
| Square Pyramid | | | | True Height | 15 | | | | | | | | | | |
| | Width | Height | | True Width: | 15 | | | | | | | | | | |
| 1 | 12.675 | 17.713 | | | | | | | | | | | | | |
| 2 | 13.162 | 20.637 | | | | | | | | | | | | | |
| 3 | 14.95 | 19.5 | | | | | | | | | | | | | |
| 4 | 14.625 | 19.998 | | | | | | | | | | | | | |
| 5 | 13 | 21.288 | | | | | | | | | | | | | |
| Average | 13.6824 | | | | | | | | | | | | | | |
| STDEV | 1.030369497 | 1.359759795 | | | | | | | | | | | | | |